Digital Admission Procedure Management for Post-GST Candidates

A Project Report Submitted

In partial fulfillment of the requirements for the degree of

B.Sc. Engineering in Computer Science and Engineering

Submitted by

Nafis Ahamed

Student ID: 200129

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Supervised by

Dr. Mohammad Nowsin Amin Sheikh

Assistant Professor

Department of Computer Science and Engineering

Faculty of Engineering and Technology

Jashore University of Science and Technology

Jashore-7408, Bangladesh

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This project titled **Digital Admission Procedure Management for Post-GST Candidates** submitted by **Nafis Ahamed** for final defence for the degree of B.Sc. in Computer Science and Engineering on April 2026.

Signature of the Student

Nafis Ahamed

Student ID: 200129

Signature of the Supervisor

Dr. Mohammad Nowsin Amin Sheikh

Assistant Professor

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I want to say thank you to God, who helped me finish this work.

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Nafis Ahamed
Student ID: 200129

Preface

On the eve of the beginning of civilization, human beings had been striving tirelessly to discover, explore, and innovate unknown aspects of the world. This inherent curiosity and desire for advancement have driven humanity toward continuous progress and development. Over time, human beings have not only acquired theoretical knowledge but have also emphasized the importance of practical implementation, which plays a crucial role in solving real-world problems and improving the quality of life.

In the modern era, the field of Computer Science and Engineering has emerged as one of the most dynamic and influential disciplines, contributing significantly to technological advancement and digital transformation. The philosophy of the Computer Science and Engineering department is centered on equipping students with both theoretical understanding and practical skills. To achieve this goal, the department has introduced thesis and project programs that encourage students to engage in real-life problem-solving and innovative system development.

As a part of fulfilling the academic requirements of the undergraduate program, I have undertaken a project titled Digital Admission Procedure Management for Post-GST Candidates. This project aims to develop a comprehensive web-based platform that simplifies and streamlines the admission process for undergraduate students. It focuses on efficiency, transparency, and user-friendliness, ensuring a better experience for both applicants and administrators.

Throughout the development of this project, I have conducted extensive research, analysis, and observation to understand the existing admission processes and identify their limitations. This has enabled me to design and implement a system that addresses real-world challenges effectively. The project has provided me with valuable opportunities to enhance my technical skills, problem-solving abilities, and understanding of software development methodologies.

I believe that this project will not only serve as an academic accomplishment but also contribute positively to the automation and digitalization of admission systems. The knowledge and experience gained during this work will undoubtedly play a vital role in my future career and professional development.

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List of Abbreviations

JUST	Jashore University of Science and Technology
DAPM-PGC	Digital Admission Procedure Management for Post-GST Candidates
CSE	Computer Science and Engineering
SRS	Software Requirement Specification
GST	General, Science and Technology
ERD	Entity Relationship Diagram
UI	User Interface
UX	User Experience
API	Application Programming Interface
SQL	Structured Query Language
RDBMS	Relational Database Management System
SDLC	Software Development Life Cycle
MVC	Model-View-Controller
JSON	JavaScript Object Notation
IEEE	Institute of Electrical and Electronics Engineers

Chapter 1

Introduction

1.1 Chapter Overview

The chapter highlights a brief discussion about the problem statement and the rationale behind undertaking this project. It starts by providing a brief introduction to the university admission process concerning the post-GST university admission process, wherein there are several stages of verification necessary to accomplish the entire process of admission. The rationale behind this project is based on the current state of affairs regarding the process of university admission. Presently, the admission process is carried out either manually or semi-manually, leading to inefficiencies in data management, which can be erroneous and slow, and poor communication between students and various departments within the university.Finally, the chapter states the objectives of the project.

1.2 Background

Admission of a university is an indispensable activity in every educational activity and is as old as education itself. With more students applying for admission than in previous years, admission process for admissions officers has become increasingly time-consuming. More applications resulted in heavier paperwork and processing challenges. Every admission is routed through various faculty for evaluation, and the manual admission process caused difficulty in admission, and archiving.

In order to speed up and simplify the admission process, the development of the Automated Final Admission System is to enact document redesign and an automated management program. The proposed “Digital Admission Procedure Management for Post-GST Candidates” would store, route, and retrieve prospective and current application documents. The development of science and technology has immensely contributed to the growth of computers and the internet which has adversely increased the use and need for an automated applicant’s admission system.

Universities, Colleges, Companies, organizations, etc. are seeking better ways to distribute and disseminate information throughout the world. In order to achieve this, they develop and rely on online applications, which help them input, update, process, and disseminate information. The internet has increased the use of online applications because data could be easily and readily accessed as well as documented. This project is targeted at allowing admission of applicants’ decisions to be made faster and more efficiently. It looks at improving the present traditional system of the manual process (paper-based) which is manually documented. This type of project is a hot topic in the whole world to digitalize our education system.

1.3 Motivation

1.3.1 Existing System

Many schools and universities use automated systems to manage the steps of admitting new students. These systems vary in what they can do how well they. How they are set up depending on what each school needs and what resources they have.

The current system used by JUST is called “Admission 2025” and it is available online. This system has some features but it does not cover everything needed.

There is unlimited user registration in the system, which poses a threat to data consistency. The system does not have an option for Excel file uploads. Furthermore, the system code has poor organization, with the front-end and back-end intertwined..

The system also does not properly check the information that is submitted which can make it hard to ensure that the information is correct and real.

1.3.2 Limitations of the System

The current system has issues that affect how well it works and how easy it is to use. The major limitations are highlighted below:

- One major issue is that there is no way to check if the information entered is correct. This means users can enter incomplete information without any restrictions.
- Another issue is that the system allows users to add data without limits but does not have ways to update or remove data. This can lead to data.
- The way the system is set up and the code it uses are not well organized. This makes it hard to fix grow or add features to the system in the future.
- Also people who have not been accepted can still sign up which reduces the accuracy and trustworthiness of the admission process.
- Key features like confirming admission sending automated messages and verifying documents are either missing or not fully working. This prevents the system from being a solution, for automating admissions.

1.4 Objectives

The main goal of this project is to create a web-based admission system that's easy to use and works well. This system will help with the following:

- Efficiency: We want to make the admission process automatic to save time and make it easier to handle data. The system will help reduce work.
- Accuracy: We aim to reduce mistakes by minimizing data entry. This will ensure that application data and documents are handled correctly.
- Consistency: The system will process all data and applications in the way. This will reduce the risk of data and data loss.
- Validation and Reliability: We will put in place checks to ensure only eligible candidates can register. The system will also ensure that candidates submit information.
- Performance: We will use modern technologies to make the system work better and more reliably.
- Security: We will update the systems security measures to protect user data. This will prevent people from accessing the system without permission.
- Scalability: The software will be designed in such a way that it is scalable and will have an adaptive nature to accommodate more and more users and applications. It will also allow for future additions like features, departments, and even admissions.
- Usability: The system will provide ease of use for students, staff members, and other users to perform their tasks smoothly without any complications.
- Maintainability: The software will be designed with a well-defined architecture that can make it easy to update, maintain, and enhance in the future.
- Transparency: The system will provide transparency in the process of admission. Users will always be able to see how far their admission applications have progressed.

1.5 Contributors

The project titled Digital Admission Procedure Management for Post-GST Candidates was completed by Nafis Ahamed (Student ID: 200129) in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering. This work was accomplished under the constant guidance and valuable supervision of Dr. Mohammad Nowsin Amin Sheikh, Associate Professor, Department of Computer Science and Engineering at Jashore University of Science and Technology. His continuous feedback, constructive criticisms, and insightful suggestions played a vital role in refining the project and ensuring the quality of the final product.

1.6 Chapter Organization

The current report is organized into six chapters to present all aspects of the project systematically:

- Chapter 1: Introduction: Provides the foundational information, including the background, motivation, and objectives behind the development of the system.
- Chapter 2: Software Requirement Specification: Details the SRS, covering the system overview, user classes, constraints, and both functional and non-functional requirements.
- Chapter 3: Design and Methodology: Explains the program design, methodology applied, use-case diagrams, and the database schema.
- Chapter 4: Implementation: Focuses on the technical implementation of the program and the description of various modules.
- Chapter 5: Testing: Discusses the testing strategies and quality assurance measures taken to ensure system reliability.
- Chapter 6: Deployment: Covers the final deployment process and future scope of the project.

Chapter 2

Background and Related Technologies

2.1 Chapter Overview

The process of enrollment in many public establishments, particularly those conducting admissions after GST (General, Science, and Technology), is highly sensitive, requiring a lot of data processing activities. Historically, the manual handling of these activities has been done in highly unstructured and inefficiently designed digitally assisted manual systems.

In this chapter, it is intended to describe the operation of the current admission management system, including its shortcomings and the technology environment used in the development of the current online admission system.

2.2 Traditional Admission Management Approaches

The traditional methods of admissions involve manual or semi-computerized procedures. This includes:

- Manual forms for students' admission
- Document submission via paper
- Use of spreadsheets (Excel) for recording applications
- Sending notifications via email
- Manual validation of application status by administrative or faculty staff

Problems Associated with Traditional Methods

- Possibility of errors during manual data entry
- Non-availability of centralized database management
- Inability to track application status
- Lack of any system for validating documents
- Tedious process of verifying documents

All of these shortcomings make the system inefficient and incapable of handling admissions such as Post-GST admissions.

2.3 Existing Digital Solutions for Admission Management

A number of universities have come up with half-baked solutions which provide applicants with the option to register online and fill out their details. But all such digital solutions lack the appropriate software architecture.

Common Features of These Systems

- Tightly coupled or monolithic frontend and backend codes
- Not having strict RBAC implementation
- No efficient file upload feature (especially for circulars or any sort of notices)
- Lack of proper data validation
- Inefficient database architecture
- Complete reliance on administrative authorities to keep a check on things

2.3.1 Flaws in Existing Digital Systems

Despite being digitized systems, these solutions carry with them certain limitations that include:

- Not Having Proper Software Architecture: Due to the lack of proper architecture in these systems, frontend and backend are tightly coupled together.
- Weak Data Validation Feature: It allows users to submit invalid data.
- No Efficient File Upload Management System: No provision for file uploads (e.g., circulars, notices, schedules).
- Administrative Authority in Manual Mode: All the data is accessible to faculty and dean in the absence of any RBAC policy.
- No Implementation of RBAC: Any authorized individual can perform actions freely.
- Not Being an Audit System: Nothing is logged while making changes to data entries.

2.4 Role-Based Access Control in Management Systems

Role-Based Access Control (RBAC) plays an important role as a security component in current web applications. This component allows users to have access to functionalities based on the role assigned.

Roles in Current Admission System

- Admin (system administration and configuration)
- Admissions Committee
- Faculty / Dean (approval rights limited)
- Student (application process and follow-up)

Advantages of RBAC

- Access control
- Data Security
- Prevention of data tampering
- Accountability of the system

The current system does not contain this component at all or it has been wrongly implemented.

2.5 The Strong Backend Design of Modern Systems

Modern systems should have a strong backend architecture so that it will scale and be maintainable.

Characteristics of the Backend Architecture

- Frontend vs. backend separation
- Design of RESTful API
- Logic segregation into services
- Database administration centrally

- Authentication and validation through middleware

The improvement that your system brings is:

- Backend API with Node.js + Express.js
- Use of Prisma ORM for database
- PostgreSQL database engine
- Separate frontend from backend (with Next.js)

2.6 Current Web Technologies Employed

Modern technologies employed for the creation of the proposed web app include:

Front-end

- Next.js (framework based on React)
- TypeScript (improves safety and maintainability)
- Tailwind CSS (CSS utility classes)
- ShadCN UI (modern UI components)

Back-end

- Node.js (runtime environment)
- Express.js (API framework)
- Prisma (ORM for managing databases and queries)
- Postman (for API testing and debugging)

Advantages of This Tech Stack

- Faster development process
- Scalable architecture
- Improved type safety (TypeScript)
- High-quality UI/UX design
- Effective communication between APIs

2.7 Comparative Analysis: Existing System vs Proposed System

Feature	Existing System	Proposed System
Architecture	Frontend & backend mixed	Fully separated (Next.js + Node.js API)
Data Validation	Weak or missing	Strong validation using backend rules
File Upload System	Not available	Available (notices, documents, schedules)
Role-Based Access	Poor or uncontrolled	Proper RBAC implementation
Data Management	Excel/manual tracking	Centralized database (Prisma ORM)
Security	Low	High (structured API + validation)
Maintainability	Difficult	Easy due to modular structure
Scalability	Limited	High scalability using modern stack
Notice System	Manual or external	Built-in dynamic notice system

2.8 System Features: Admin Management

2.8.1 Description and Priority

The Admin module has the highest priority and controls the entire system. The admin uploads applicant data (Excel file), manages users, controls admission deadlines, and monitors all activities.

2.8.2 Functional Requirements

Authentication: Facilitates the login process for users via their username and password. This feature guarantees secure login and prevents unauthorized usage of the system.

- Input: Username and password
- Output: Secure login access

Upload Applicant Data: Facilitates the uploading of applicant data into the system via Excel sheets. The system automatically generates accounts for all applicants after the data importing process is complete.

- Input: Excel file containing applicant information
- Output: System generates applicant accounts

Manage Users: Facilitates the management of user accounts by allowing administrators to view, update, and delete accounts as needed. This fosters effective user control and comprehensive management of the system environment.

- View, update, and delete users

Update Applicant Information: Facilitates the modification of applicant data whenever changes are required, ensuring that all stored information remains accurate and up-to-date.

- Modify applicant data if required

Control Admission: Facilitates the opening and closing of the admission process for candidates. This assists in controlling the application timeline and managing system availability.

- Open or close admission system

View Reports: Generates detailed reports regarding the admission process and seat availability. This enables the administrator to monitor progress and maintain oversight of the entire system.

- Monitor admission progress and seat status

2.9 System Features: Dean Office

2.9.1 Description

The Dean Office verifies applicant information and approves or rejects applications based on eligibility criteria.

2.9.2 Functional Requirements

- Search applicant using ID
- View applicant profile
- Verify documents
- Approve or reject applicant
- View applicant lists (all, pending, approved)

2.10 System Features: Hall Office

2.10.1 Description

The Hall Office Module is in charge of handling financial and accommodation clearances of the applicants. It checks the validity of the applicants' payments using their bank statements and makes sure that all necessary payments are made correctly. The module also takes care of accommodation clearances of qualified students.

2.10.2 Functional Requirements

- Search applicant
- View profile
- Verify bank receipt
- Approve applicant

2.11 System Features: Medical Office

2.11.1 Description

Task of conducting medical evaluations for the applicants falls to the Medical Office module. Medical data are collected, and required physical examinations are performed. On completion of the procedure, medical report cards are produced for each applicant. This guarantees selection of medically sound applicants.

2.11.2 Functional Requirements

- Search applicant
- Insert medical data
- Generate report
- Approve applicant

2.12 System Features: Registrar Office

2.12.1 Description

The function of the Registrar Office module lies in granting final approval to applicants admitted. It checks all documents submitted by the applicant and ensures that all previous stages of approvals are done. Also, it keeps the official record of the student.

2.12.2 Functional Requirements

- Search applicant
- Receive documents
- Final approval
- Generate reports

2.13 Non-Functional Requirements

2.13.1 Performance

The system should be able to handle many users and queries simultaneously, which would enable quick responses during the admission processes. It would be required to optimize queries to the database and improve server performance for ensuring seamless system operation.

2.13.2 Security

The system must protect information from any external threats by incorporating proper authentication mechanisms and role-based access control methods. Information encryption and other security protocols are necessary for achieving security objectives by keeping user information strictly confidential.

2.13.3 Scalability

The system must be scalable enough to support more users in the future, ensuring that the system can be scaled up easily according to its capabilities. The flexibility of designing and allocating resources is critical for facilitating future usability and stability of the system.

2.13.4 Maintainability

A good code structure is needed for facilitating modifications and debugging of the system. Documentation plays an important role in supporting maintainability goals by enabling developers to perform necessary changes quickly without raising expenses.

2.13.5 Usability

A user-friendly interface must be made available to all users and must ensure that there is no confusion in using the system as everything will be done in an efficient manner. Tasks must be performed by users easily and with clarity.

2.13.6 Reliability

The operation of the system must never fail, and availability must be assured at all times. The correct handling of errors must be ensured through the use of backups.

2.13.7 Availability

The system must always be available to the user. Proper management of server uptime and monitoring processes will help ensure the availability of the system. Load balancing and failover capabilities can be deployed to guarantee constant availability of the system.

2.13.8 Flexibility

The system must be flexible enough to adjust to any changes in the policies, procedures, or even the rules of admission. The system should be capable of updating itself based on the admission process without undergoing extensive development activities. Flexibility is important for ensuring that the system retains its relevance.

Chapter 3

System Design and Methodology

3.1 Chapter Overview

This chapter gives an insight into the architecture of the system design before the actual implementation of the system in question. Starting off, use case diagrams show how various people interact with the system, and the chapter breaks down particular parts of the process involved like secure user logins, submission of application forms by the applicant, and the verification process by the Dean. Through these, the processes of approval of each office including the Hall, Medical and Registrar's office are depicted to give an overview of the sequence of processing admission data in various departments.

The chapter further provides a schematic diagram of the system's database design. Various tables of data, for example, for users, applicants, and other supporting document information are discussed in detail. There is an emphasis on various relationships that exist between various types of data in the system. The data flow architecture is included in the discussions to give technical details on how the data flows within the environment.

Aside from the structure and data-related perspectives on the design, another important perspective that is emphasized in this chapter is the modular design of the system. Every functional requirement is designed as an individual module to allow further improvements in the future without disrupting the entire functioning of the system. A significant part of designing the modules will be to incorporate security measures like user authentication, authorization, and data validation to ensure that sensitive admission data is secure against any threat. Besides, it is necessary to explain the integration of the back-end functionalities with the graphical user interface.

Aside from the structure and data-related perspectives on the design, another important perspective that is emphasized in this chapter is the modular design of the system. Every functional requirement is designed as an individual module to allow further improvements in the future without disrupting the entire functioning of the system. A significant part of designing the modules will be to incorporate security measures like user authentication, authorization, and data validation to ensure that sensitive admission data is secure against any threat. Besides, it is necessary to explain the integration of the back-end functionalities with the graphical user interface.

Additionally to what was already stated regarding the architecture design and data storage capabilities, this chapter emphasizes the necessity of creating a coherent and component-oriented system design. Every functional requirement is realized as a separate and independent module, making the system flexible and scalable in terms of further implementation.

3.2 Use Case Diagram

3.2.1 User Management

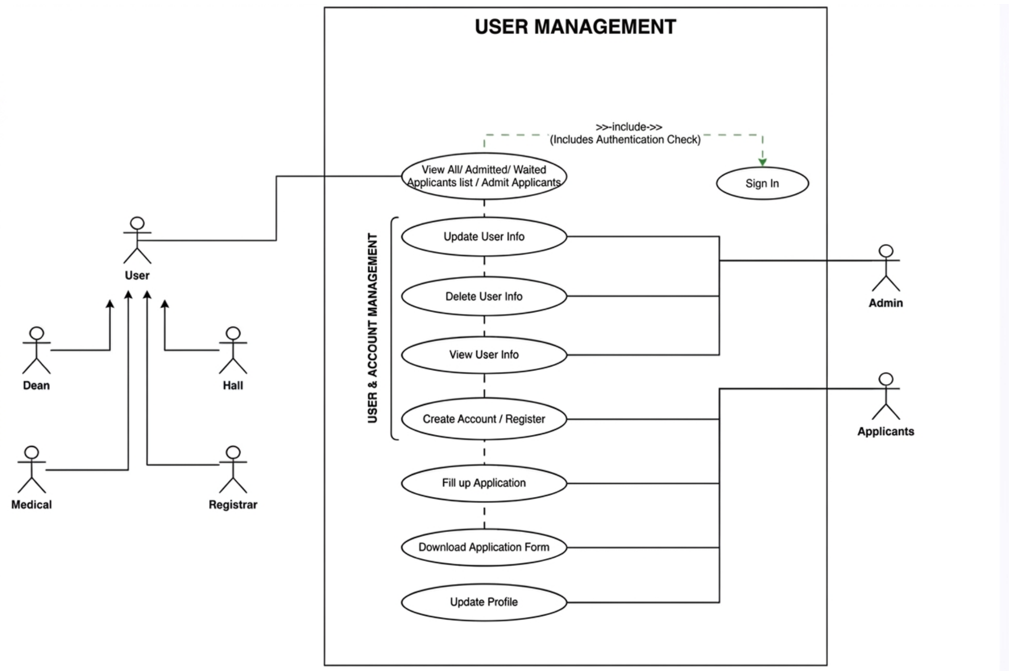


Figure 3.1: Use Case Diagram of User Management

The User Management use case diagram shows how the various participants interface with the system as per their different responsibilities. The is the main external participant who starts the process of applying for admission to the university by entering the system and filling the form containing personal and academic data. From the administration point of view,the is a responsible for managing the user accounts, accessing control of the system, and granting authorities for all offices to operate. The Admin also ensures that all system actions are consistent and safe. All participants perform their interaction with the system by proper authentication processes.

The diagram of the user management use case depicts interactions between various stakeholders based on the roles that they play. The is an important stakeholder because he/she triggers the admission process by logging into the system and filling out the application form with relevant details. In terms of system administration, it is the job to ensure that there is efficient management of user accounts as well as system access. The role of the Admin also includes monitoring and maintenance of system activities.

3.2.1.1 Login Process

This use case describes the method used to authenticate access to the Digital Admission Procedure Management system for post-GST candidates. This process helps secure the system by allowing only authorized users to have access to certain functionalities of the system. In order to keep the system safe from any unauthorized access, the system's login capability is totally dependent upon the availability of user records; in other words, if there are no user details imported using an Excel file, login will be restricted..

Table 3.1: Use Case Table: Login Process (UC-1.1)

Characteristics	Description
Use Case ID	UC-1.1
Use Case Name	Login Process
Primary Actors	Admin, Dean Office, Hall Office, Medical Office, Registrar Office, Applicant
Purpose	To allow users to securely access the system dashboard.
Precondition	The user must have a valid account with a username or email and password.
Post-condition	The user is redirected to their respective dashboard based on their role.
Trigger	The user tries to access a protected page or clicks on the login option.
Scenario	1. The user opens the login page of the system. 2. The user enters their username or email along with the password. 3. The user clicks the Login button. 4. The system checks the entered information with the PostgreSQL database using Next.js and Prisma. 5. If the information is correct, a secure session is created. 6. The user is then allowed to enter the dashboard.
Exception	Invalid Credentials: If the entered information is incorrect, the system shows an "Access Denied" message.
Frequency of Use	Very High (used daily).

3.2.1.2 View Applicant List

This use case allows administrative staff to access and browse the list of applicants for admission. It provides Admin, Dean, Hall, Medical, and Registrar offices with the ability to filter and monitor applicants based on their category.

Table 3.2: Use Case Table: View Applicant List (UC-1.2)

Characteristics	Description
Use Case ID	UC-1.2
Use Case Name	View Applicant List
Primary Actors	Admin, Dean Office, Hall Office, Medical Office, Registrar Office
Goal	To sort and display applicant records by filtering categories such as All, Admitted, or Waiting.
Precondition	The user must be logged in with proper credentials to view applicant records.
Post-condition	The selected applicant list is displayed on the user interface.
Trigger	When there is a need to verify an applicant's status or departmental clearance.
Scenario	1. The user logs in and opens the Dashboard. 2. The user selects 'Applicant Management' or 'View List' from the sidebar menu. 3. The user chooses a specific category (All Applicants, Admitted, or Waiting). 4. The system retrieves the relevant data from the PostgreSQL database using Prisma. 5. The applicant records are displayed in a structured table for review.
Exception	Access Denied: If the user does not have permission to view the list, an "Unauthorized Access" message is shown.
Frequency of Use	High (used frequently by administrative staff).

3.2.1.3 Update User Information

This use case outlines the procedure of changing user information within the system to ensure that there is accurate information about all users in the *Digital Admission Procedure Management for Post-GST Candidates* as an administrator.

Table 3.3: Use Case Table: Update User Information (UC-1.3)

Characteristics	Description
Use Case ID	UC-1.3
Use Case Name	Update User Information
Primary Actors	Admin
Goal in Context	To change or update user information as needed.
Precondition	Admin is successfully signed in into the administrative dashboard.
Post-condition	The user data is correctly updated within the PostgreSQL database.
Trigger	A need to update or correct any information regarding a user in the system.
Scenario	1. The Admin opens the User Management tab in the admin dashboard. 2. The Admin finds the specific user in question and clicks on the Edit button/icon. 3. The Admin receives the user's data in a filled form from the database. 4. The necessary information fields, such as name, email, role, or status, are edited by the Admin. 5. The Admin clicks on Update or Save Changes. 6. The Next.js backend validates and saves the new user data using Prisma. 7. The Admin receives a success message.
Exception	Validation Error: In case of entering invalid data such as a duplicate email address, the system prevents the update and shows an error message.
Frequency of Use	High (Used regularly for maintaining user data).

3.2.1.4 View User Information

This particular use case will allow the Admin to see the total information regarding profiles and access levels of all the users (Applicants and Staff) from the system *Digital Admission Procedure Management for Post-GST Candidates*. Using this use case, the Admin would be able to manage the information of all the users, like personal data, user roles, and current status in the system. With the help of this use case, the Admin would be able to monitor the activities of the users and make sure that all the users are working on their permissible levels. The Admin would even be able to manage the inactive and unauthorized users and take appropriate steps, which may include updating access, limiting access, or deleting access of these users. This will not only improve security but also prevent any kind of abuse of critical information relating to the admission procedure. Moreover, this use case also helps in improving transparency by providing information about all the users of the system.

Table 3.4: Use Case Table: View User Information (UC-1.5)

Characteristics	Description
Use Case ID	UC-1.5
Use Case Name	View User Information
Primary Actors	Admin
Goal in Context	To display the complete profile details, account status, and role information of a particular user.
Precondition	The Admin is required to be authenticated and logged into the administrative dashboard.
Post-condition	The detailed information of a particular user is obtained from the database and shown on the screen.
Trigger	The Admin has a need to verify the credentials of the user or investigate a particular account.
Scenario	1. The Admin accesses the User Management section or all users in the dashboard. 2. The Admin searches for the desired user using their ID, Name, or Email using the search bar provided. 3. The Admin selects the 'View Details' or 'Profile' button of the user. 4. The Next.js application queries for that particular user's information stored in the PostgreSQL database using Prisma. 5. The user's profile is rendered on the screen.
Exception	User Not Found: If the requested user ID does not exist, the system displays a "Record Not Found" message.
Frequency of Use	High (Used frequently for administrative oversight).

3.2.1.5 Fill-Up Application Form

This use case will describe how the process of filling out the application form using the *Digital Admission Procedure Management for Post-GST Candidates* works. This process involves several steps in order to obtain the documentation and other important information needed for the final admission validation.

Table 3.5: Use Case Table: Fill-Up Application Form (UC-1.6)

Characteristics	Description
Use Case ID	UC-1.6
Use Case Name	Fill-Up Application Form
Primary Actors	Applicant
Goal in Context	To successfully submit the completed admission application form.
Precondition	The user needs to sign in and should not have a pending application already submitted.
Post-condition	The admission application form is saved in the database under the ”Pending” status.
Triggers	Clicking ”Apply now” or ”Fill-up Form” from the dashboard menu.
Scenario	1. The applicant goes to the admission form section. 2. The applicant sees that the form consists of several stages (Personal Information, Academic History, Document Upload). 3. The applicant fills in the form with the needed data (Photo, Signature, etc.). 4. The applicant checks if the entered data is correct. 5. The applicant clicks the Submit button. 6. The Next.js frontend passes the information to the API, then it is saved to PostgreSQL via Prisma.
Exception	Validation Error: If required fields are missing or uploaded files are in unsupported formats, the system prompts the applicant to correct the errors.
Frequency of Use	Once per applicant (High during admission periods).

3.2.1.6 Download Application Form

Through this use case, the applicant will be able to generate and download the PDF copy of the application form he/she has applied for. The use case acts as an official record for the applicant and plays a very important role in the physical verification process conducted at the *JUST*. The generated PDF file contains all the information regarding the application form along with the personal details of the applicant, including his/her educational qualifications and application status. Through this feature, applicants can download and print the document, thus minimizing any need for manual work and the possibility of any information getting lost in between. It not only increases the efficiency of the entire system but also enhances user-friendliness of the system.

Table 3.6: Use Case Table: Download Application Form (UC-1.7)

Characteristics	Description
Use Case ID	UC-1.7
Use Case Name	Download Application Form
Primary Actors	Applicant
Goal in Context	To have a digital/physical copy of their admission application form as documentation.
Precondition	The applicant should have logged in to the application and should have submitted his/her application form already.
Post-condition	A PDF file will be generated and downloaded to the applicant's device.
Triggers	The applicant clicks the "Download PDF" or "Print Application" button in their dashboard.
Scenario	1. The applicant logs into their personal dashboard. 2. The system checks the application status from the database using PostgreSQL. 3. The applicant presses the button "Download PDF" on the dashboard. 4. The server-side of the Next.js generates the application in PDF format based on the stored application data. 5. The file download starts in the browser of the applicant. 6. The applicant downloads the application form in PDF format.
Exception	Generation Error: If the server fails or the database cannot be reached, the system displays a "PDF Generation Failed" error.
Frequency of Use	High (Once per applicant after submission).

3.2.2 Dean Office Management

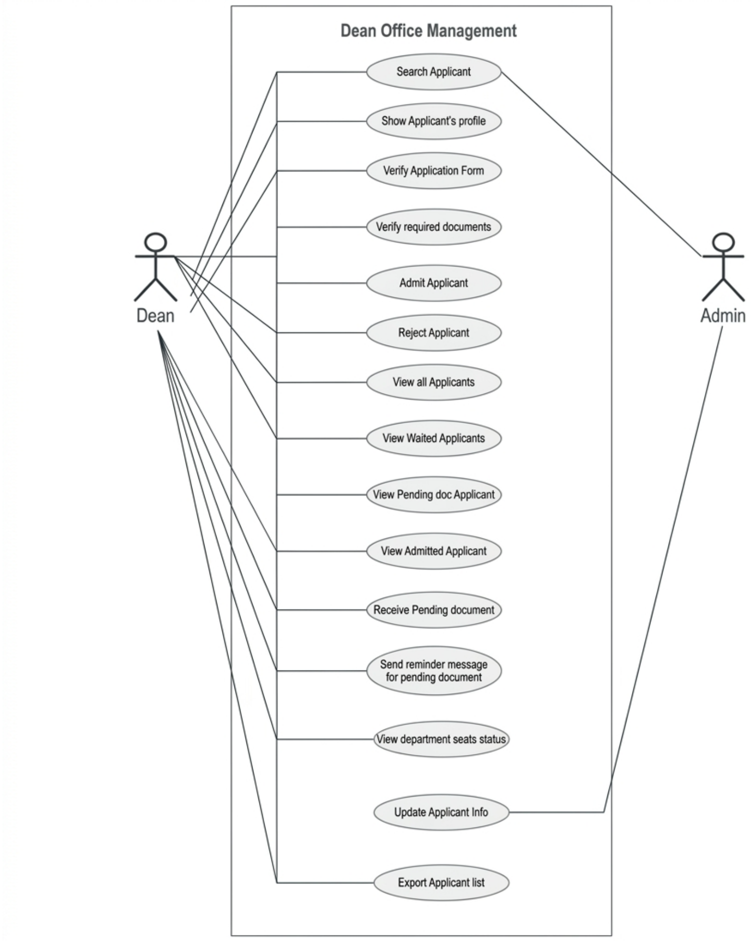


Figure 3.2: Use Case Diagram of Dean Office Management

The Dean Office Management use case diagram consists of two primary actors: Admin and Dean Office. This module controls various types of use cases including searching applicants, viewing applicant profiles, verifying application forms, verifying required documents, admitting applicants, rejecting applicants, and managing applicant lists. It also provides functionalities to view all applicants, waited applicants, and admitted applicants with or without pending documents.

3.2.2.1 Applicant Search

In this use case, you will learn about how the Dean Office uses the system to find a particular applicant. This makes it possible for those authorized to find applicant records through the search parameter like application number, name, or registration number. This feature ensures that there is a reduced workload on the side of the Dean Office whenever it involves searching for numerous admission records. By applying this functionality, the

Dean Office can view all academic and personal details entered during the application process. It should be noted that there is a provision for sorting and filtering results from the search. This makes it possible to view accurate results.

Table 3.7: Use Case Table: Applicant Search (UC-2.1)

Characteristics	Description
Use Case ID	UC-2.1
Use Case Name	Applicant Search
Actors	Dean Office
Goal Statement	To conduct a search for a specific applicant using Applicant ID or Roll.
Preconditions	The user must be logged in and have the Applicant ID or Roll.
Trigger	The need to find information about a particular applicant.
Steps	1. Navigate to the Dean Office dashboard. 2. Enter the Applicant ID or Roll in the search field. 3. Press Enter to perform the search.
Exceptions	Invalid role.
Occurrence	Very frequent

3.2.2.2 Display Applicant Profile

This use case explains the Dean Office's perspective on the profile of the applicant.The profile contains all the details about the applicant including personal as well as academic information. The reason behind including these elements is that it makes it easy to admit the students. Apart from these elements, the system includes the documentary evidence, standard test scores, and academic records of the candidate. In short, all of the elements are provided to the user in one central location, where he can easily assess whether the person deserves to be admitted or not. Moreover, the system highlights those elements in the data that are inconsistent,and there are quick action buttons as well, which facilitate the process of taking.the final decision regarding the candidate.In addition to the above benefits, the system also offers a well-structured format of presenting the applicant's information, thus facilitating the work of the Dean's Office by enabling easy navigation between various sections. Moreover, the system allows tracking the modification history, which means that it is possible to identify all changes that have been introduced to an applicant's data at some point of time. Validation flags assist in quickly finding out whether there is any missing information. Finally, the system guarantees quick and easy access to all uploaded documents. Thus, the probability of any mistakes during evaluating applicants' data will be substantially reduced.

Table 3.8: Use Case Table: Display Applicant Profile (UC-2.2)

Features	Description
Use Case ID	UC-2.2
Use Case Name	Display Applicant Profile
Actors	Dean Office
Goal Statement	To display the detailed profile of the selected applicant for assessment purposes.
Precondition	The user should be logged in and must have searched for the applicant.
Postcondition	The applicant data will be shown on the screen from the PostgreSQL database using Prisma ORM.
Trigger	When there is a need to view personal or academic information of an applicant during the admission process.
Steps	1. Search for a particular applicant from the applicant list. 2. Click on the Profile or Details button. 3. The system retrieves the applicant's data. 4. The system displays the applicant information on the screen.
Exception	Data not available: An error message will be shown. Access forbidden: The user cannot view the data.
Frequency	High (When the admission process is underway)

3.2.2.3 Validate Applicant's Form

This use case describes the process through which the Dean Office verifies the application form of the applicant. It makes sure that the details submitted are accurate before granting final permission. At this phase, the system automatically compares the details entered in the form with the electronic records uploaded to detect any discrepancies. The Dean Office examines essential elements, such as candidate qualification, quota availability, and GST test results to safeguard the entire admission process. When the verification process is complete successfully, the application is tagged as "Verified," and the candidate advances to the next level of registration. In cases where there are discrepancies, the Dean Office may mark the application for corrections or explain why it was rejected.

Table 3.9: Use Case Table: Validate Applicant's Form (UC-2.3)

Characteristics	Description
Use Case ID	UC-2.3
Use Case Name	Validate Applicant's Form
Primary Actors	Dean Office
Goal in Context	Validation of the applicant's application form.
Pre-condition	The user must log in as Dean Office and an application must be pending for verification.
Post Condition	The application status is updated as 'Verified' in the PostgreSQL database.
Triggers	Processing and validation of applications for the final admission list.
Scenario	1. The user accesses the pending application records. 2. The user navigates to the Verification or Review page. 3. The system displays the submitted form data. 4. The user checks the data and clicks Approve or Verify. 5. The system updates the status using Prisma ORM.
Exception	Data Mismatch: If the data does not match, the user may reject the form or ask the applicant to resubmit it.
Frequency of Use	High (in peak admission periods)

3.2.2.4 Admit Applicant (Dean Clearance)

The use case gives the Dean Office the ability to provide final clearance for admission after successful verification of all records.Furthermore, the system ensures that the final clearance is only provided if allnecessary verification steps have been completed successfully. This allows the Dean Office to see a comprehensive report on the applicant's academic, medical, and administrative records, before issuing its final approval. It also includes an interface which helps in either accepting or rejecting the decision along with proper justification. As soon as the clearance is provided, the system makes automatic changes to the status of the applicant in all the modules. This means that the applicant will not be able to make any further changes to the approved application without valid permission from relevant authorities. In addition, it creates an entry for final approval so as to keep record for future references. It also sends notification to the applicant, once the final decision is made. The proposed work flow prevents applicants from skipping any step during their verification process. Also, there is no need for manual verification as the system automatically makes changes in other modules.

Table 3.10: Use Case Table 2.12: UC-2.5

Characteristics	Description
Use Case ID	UC-2.5
Name of Use Case	Admit Applicant
Main Actors	Dean Office
Goal	The goal of the use case is to allow admission clearance of the applicant.
Precondition	Documents should be verified, and the applicant should be eligible.
Post Condition	The applicant is set as “Admitted,” while the seats in the department are increased.
Steps	1. Find the eligible record of the applicant. 2. Read the final summary of the candidate. 3. Press the Admit button. 4. Record is saved in PostgreSQL database.
Exceptions	Full seat or unauthorized role.
Usage Frequency	Very Frequent

3.2.2.5 Application List Views (All, Waiting, Admittees)

This use case describes the procedure for how the Dean’s office is able to monitor different kinds of applicants using the sidebar menu in the dashboard.The system offers navigation categories that can be used to screen applicants depending on their academic standing and documents submitted.The Dean’s office members can conveniently switch between different applicants for the sake of efficiency.Each category shows all the necessary information about the respective applicants in an orderly manner.The dashboard guarantees instant update of the system such that changes in the application standing will always be seen by the Dean’s office.It also makes it easier for the Dean’s office to track pending, accepted, and denied applications independently.In this way, the verification process becomes more effective.Moreover, the sidebar navigation in the system is created in a way that allows users to switch to various sections effortlessly. The system sorts applicants according to several parameters, including verification status, completeness of submitted documents, and whether there are requirements to fulfill for particular applicants. It will assist in minimizing confusion while working with numerous applications. There are options for fast search and filtration of applicants in each section. Visual markers such as statuses and color codes can be used to indicate the current status of each application, which will make it easier for the Dean's office to see applications that need more attention.The system includes the real-time synchronization function so that changes made in applications by other offices are immediately visible from the dashboard. It helps avoid misunderstandings and miscommunications between offices as well as inconsistency. In addition, the sidebar navigation makes it possible to easily switch between various workflow stages to increase overall productivity. User activity can be recorded in the system so that it can be used to track the decision-making process later on.

Table 3.11: Use Case Table: Application List Management (UC-2.7 - 2.9)

Characteristic	Explanation
Use Case Number	UC-2.7, 2.8, 2.9
Name of Use Case	View All, Waiting & Admitted Applicants
Actors	Dean's Office
Steps	1. Login to the Dean's Dashboard. 2. Select the specific type (All / Waiting / Admitted). 3. The system will then search for relevant information in the PostgreSQL database through Prisma.
Frequency of Use	High

3.2.2.6 Pending Documents and Reminders

This use case involves collecting pending documents and issuing reminders to those applicants who have yet to provide complete documents. The application will identify those applicants that are lacking or having incomplete documentation in the process of verification. An automatic list of applicants with incomplete documentation will be created by the system. The reminders could be issued to the concerned applicants using the system's interface by authorized personnel. Through reminders, the applicant will come to know which documents he/she needs to supply in order to complete his/her application. The reminders could also keep a record as to whether the applicants responded to the reminders or not.

Table 3.12: Use Case Table: Pending Documents Management (UC-2.11 - 2.13)

Characteristic	Description
Use Case ID	UC-2.11, 2.12, 2.13
Use Case Name	Manage Pending Documents & Send Reminders
Primary Actor	Dean Office
Reminders Reminder Scenario:	1. Locate the applicants whose documents are incomplete. 2. Identify the particular incomplete documents for each applicant. 3. Pick the necessary documents and click on Send Message. 4. A notification alert will be sent by the system to the applicant. 5. Reminder status will be updated once notification alert is sent. 6. The system will save the reminder history for any future use. 7. The dean office can re-notify the applicant if the application does not respond to the notification.
Receive Documents:	1. Open the Action Pending Documents on the side bar menu. 2. Crosscheck the submitted physical documents. 3. Update the status of the documents that were successfully checked. 4. Register the receipt of the missing documents. 5. Update the verification status of the applicant automatically. 6. Save the updated information for auditing purposes.

3.2.3 Hall Office Management

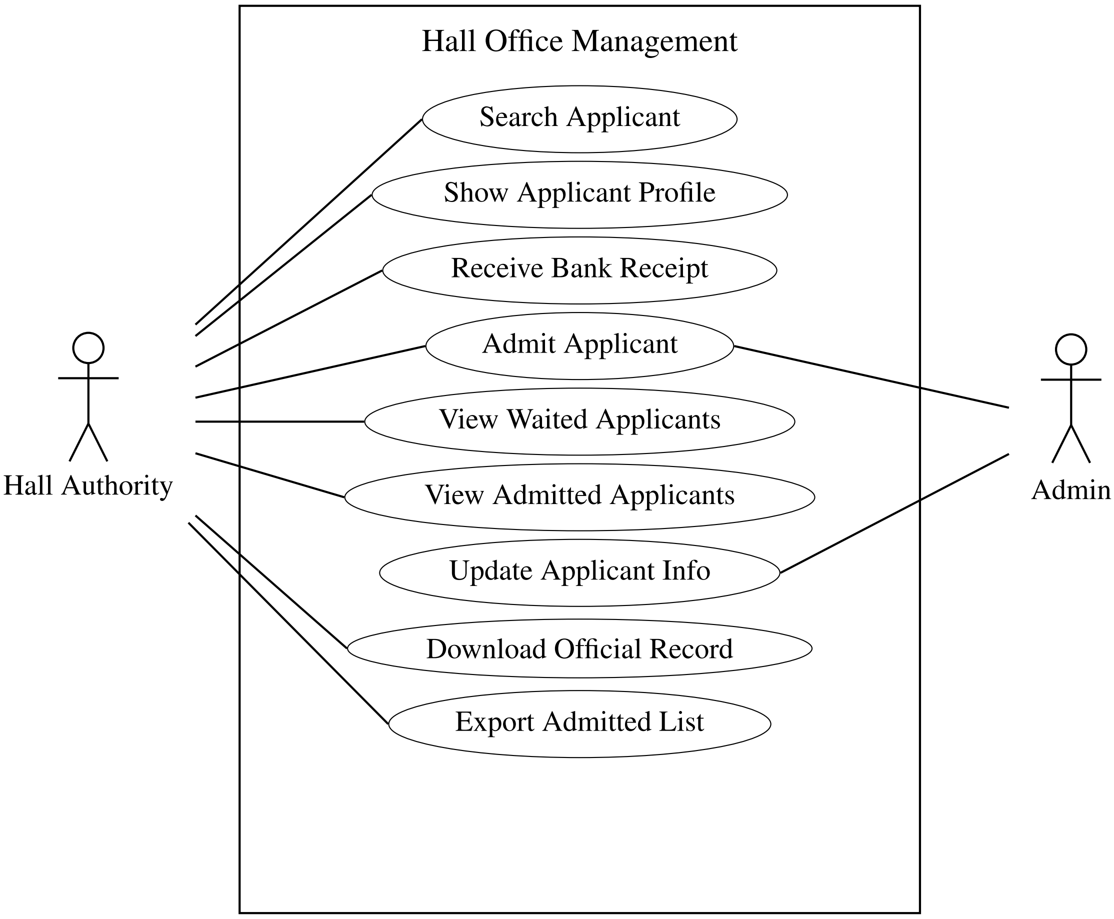


Figure 3.3: Use Case Diagram of Hall Office Management

The Hall Office Management use case diagram consists of two primary actors: Admin and Hall Authority. This module handles various operations including searching applicants, admitting them after bank receipt verification, managing lists, and updating information.

3.2.3.1 Search for Applicant

In this use case, the Hall Office will be able to conduct searches for applicant data using either the GST ID or the roll number of the applicant. This feature makes it easy for one to be able to get the applicant's data within a short time.Moreover, the system helps display the results in a systematic way, which allows the Hall Office to easily find the right candidate from amongst all those who have applied. The system makes the process of data search much easier and faster for the Hall Office. The search feature also helps minimize any risk of errors as the records can be easily identified using GST ID or roll numbers. Another advantage of the search functionality of the system is that it gives the Hall Office immediate access to all the information about each and every applicant.

Table 3.13: Use Case Table: Search Applicant (UC-3.1)

Description of Characteristics	Details
Use Case ID	UC-3.1
Use Case Name	Search Applicant
Actors	Hall Office
Purpose	To locate an applicant's details in the system.
Pre-condition	The user must be logged into the system and have the GST Applicant ID or Roll.
Scenarios	1. User navigates to the Hall Office dashboard. 2. User enters the GST ID or Roll Number. 3. System fetches and displays the applicant details.
Occurrence	Often

3.2.3.2 Receive Bank Receipt

This use case enables the Hall Office to collect and validate the applicant's bank receipt for their hall fees. The purpose is to ensure that the payment has been made either manually or digitally. Once the receipt is validated, the applicant's payment status will be updated in the system.

Table 3.14: Use Case: Receive Bank Receipt (UC-3.3)

Characteristics	Description
Use Case ID	UC-3.3
Use Case Name	Receive Bank Receipt
Primary Actors	Hall Office
Goal	To confirm the physical or electronic copy of hall fees.
Scenario	1. Find the applicant. 2. Click 'Receive Receipt' button. 3. The system will update PostgreSQL database using Prisma.

3.2.3.3 Admit Applicant (Hall Clearance)

The use case enables the Hall Office to allow the candidate to join the hall once their payment proof is verified. This use case makes sure that all the necessary requirements have been met before allowing the candidate into the hall. The admission process will be complete once the “Admit” button is pressed.

Table 3.15: Use Case: Admit Applicant (UC-3.4)

Characteristics	Description
Use Case ID	UC-3.4
Use Case Name	Admit Applicant (Hall Clearance)
Primary Actors	Hall Office
Goal	To admit the applicant.
Scenario	1. Confirm receipt. 2. Click ’Admit’ button.

3.2.3.4 Viewing Lists of Candidates

This use case is useful for the Hall Administration because it enables the administrators to see lists of candidates who have waited and been admitted. The use case helps in keeping tabs on the status of the applicants at various stages of the admission process. The user is able to access the lists via the sidebar menu.

Table 3.16: Use Case: Candidate Lists (UC-3.5 – 3.6)

Attributes	Description
Use Case ID	UC-3.5, 3.6
Use Case Title	View Waited / Admitted Candidates
Primary Actor	Hall Administration
Objective	Track the status of candidates in various stages.
Scenarios	1. Move to the sidebar menu. 2. Select the relevant list (Waited/Admitted).

3.2.3.5 Exporting and Downloading Candidate Records

This case study gives users the ability to export/download candidate data using various formats, including PDF or CSV. The functionality makes it very easy for users to create reports to document their findings. Users have direct access to the feature from the candidate data page. It takes a few seconds for the system to prepare the download file.

Table 3.17: Use Case: Candidate Report Generation (UC-3.8 – 3.9)

Attributes	Description
Use Case ID	UC-3.8, 3.9
Use Case Title	Export List / Download Official Candidate Record
Primary Actor	Hall Administration
Objective	Create PDF or CSV files of office candidate records.
Scenarios	1. Move to the list. 2. Click Export/PDF.

3.2.4 Medical Office Management

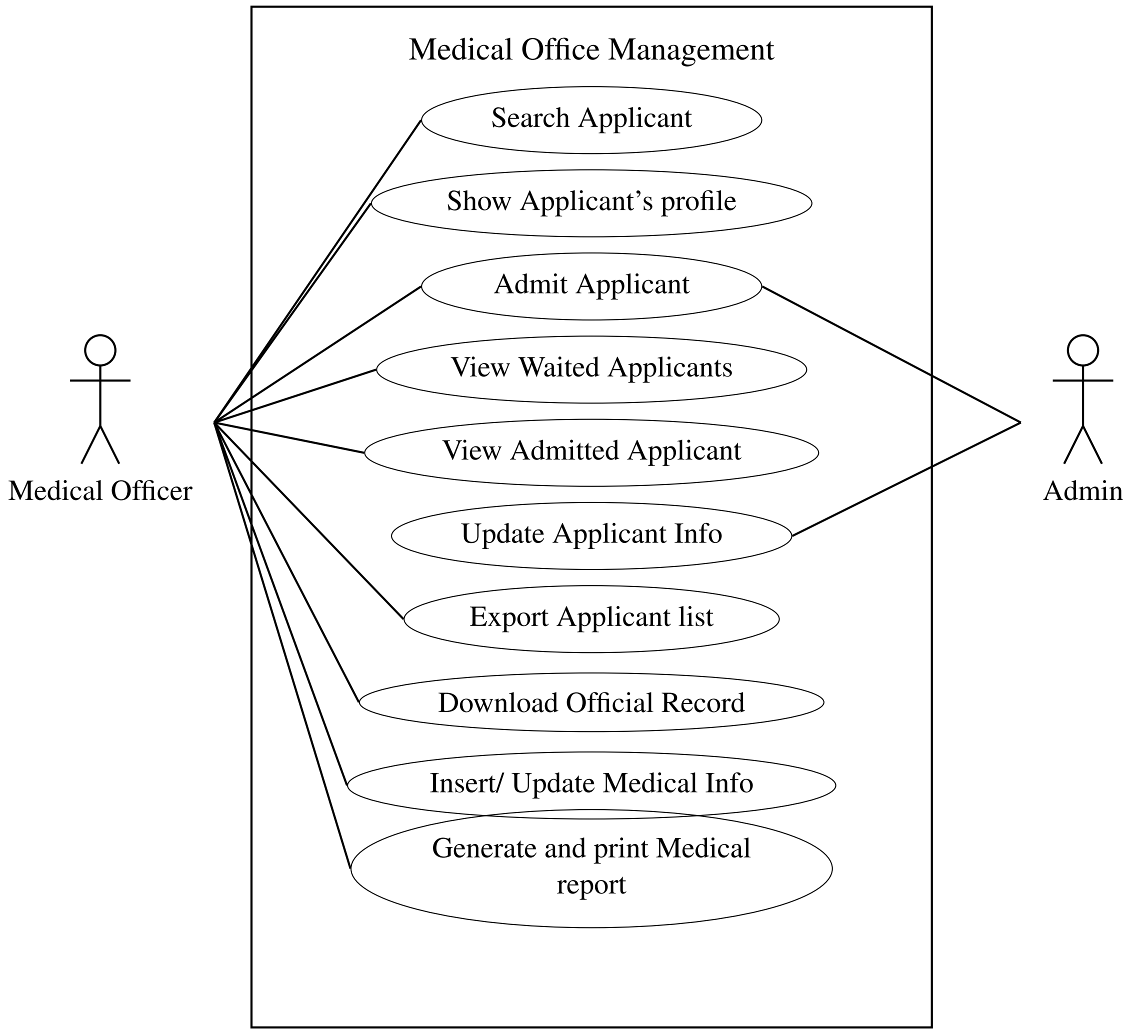


Figure 3.4: Use Case Diagram of Medical Office Management

The Medical Office Management module handles the physical examination and medical clearance of applicants. This use case diagram involves two primary actors: Medical Officer and Admin. The system allows managing medical records, generating reports, and granting final medical clearance.

3.2.4.1 Show Applicant Profile

The purpose of this use case is to enable the Medical Office to be able to see the full details of the applicant once he or she has been searched. It offers vital information about the person, including personal and medical-related information. The profile is easily accessed through the click of the "Profile" button.

Table 3.18: Use Case Table: Show Applicant Profile (UC-4.2)

Characteristics	Description
Use Case ID	UC-4.2
Use Case Name	Show Applicant Profile
Primary Actors	Medical Office
Scenario	1. Search Applicant. 2. Click Profile Button.

3.2.4.2 Insert/Update Medical Information

This particular case enables the Medical Office to input or modify the details related to the medical history of an applicant into the system. It will help in assessing the physical fitness of the applicant by making entries in the correct manner. In order to do this, the concerned medical office can complete the necessary form after finding the particular applicant.

Table 3.19: Use Case Table: Insert/Update Medical Information (UC-4.3)

Characteristics	Description
Use Case ID	UC-4.3
Use Case Name	Insert/Update Medical Information
Primary Actors	Medical Office
Goal	To Evaluate Applicant's physical state and capture information.
Scenario	1. Search Applicant. 2. Fill out Medical Info Form. 3. Submit.

3.2.4.3 Generate and Print Medical Report

The use case makes it possible for the Medical Office to create and print a medical report of the applicant. This is a detailed report that summarizes the medical assessment made by the organization. After locating the applicant, one can create his/her report using a single click. The application will then prepare the report for either printing or downloading.

Table 3.20: Use Case Table: Generate and Print Medical Report (UC-4.4)

Characteristics	Description
Use Case ID	UC-4.4
Use Case Name	Generate and Print Medical Report
Primary Actors	Medical Office
Scenario	1. Search Applicant. 2. Click Generate and Print Button.

3.2.4.4 Admit Applicant (Medical Clearance)

The medical use case will enable the Medical Office to provide a clearance for admission once he finishes carrying out the health checks. This will allow one to be sure that the applicant is medically fit based on the examination results. Once one finishes searching for the applicant, he can approve his admission by clicking the ‘admit’ icon.

Table 3.21: Use Case Table: Admit Applicant (UC-4.5)

Characteristics	Description
Use Case ID	UC-4.5
Use Case Name	Admit Applicant
Primary Actors	Medical Office
Goal	Obtain clearance following medical check-up.
Scenario	1. Look up applicant. 2. Hit admit button.

3.2.5 Registrar Office Management

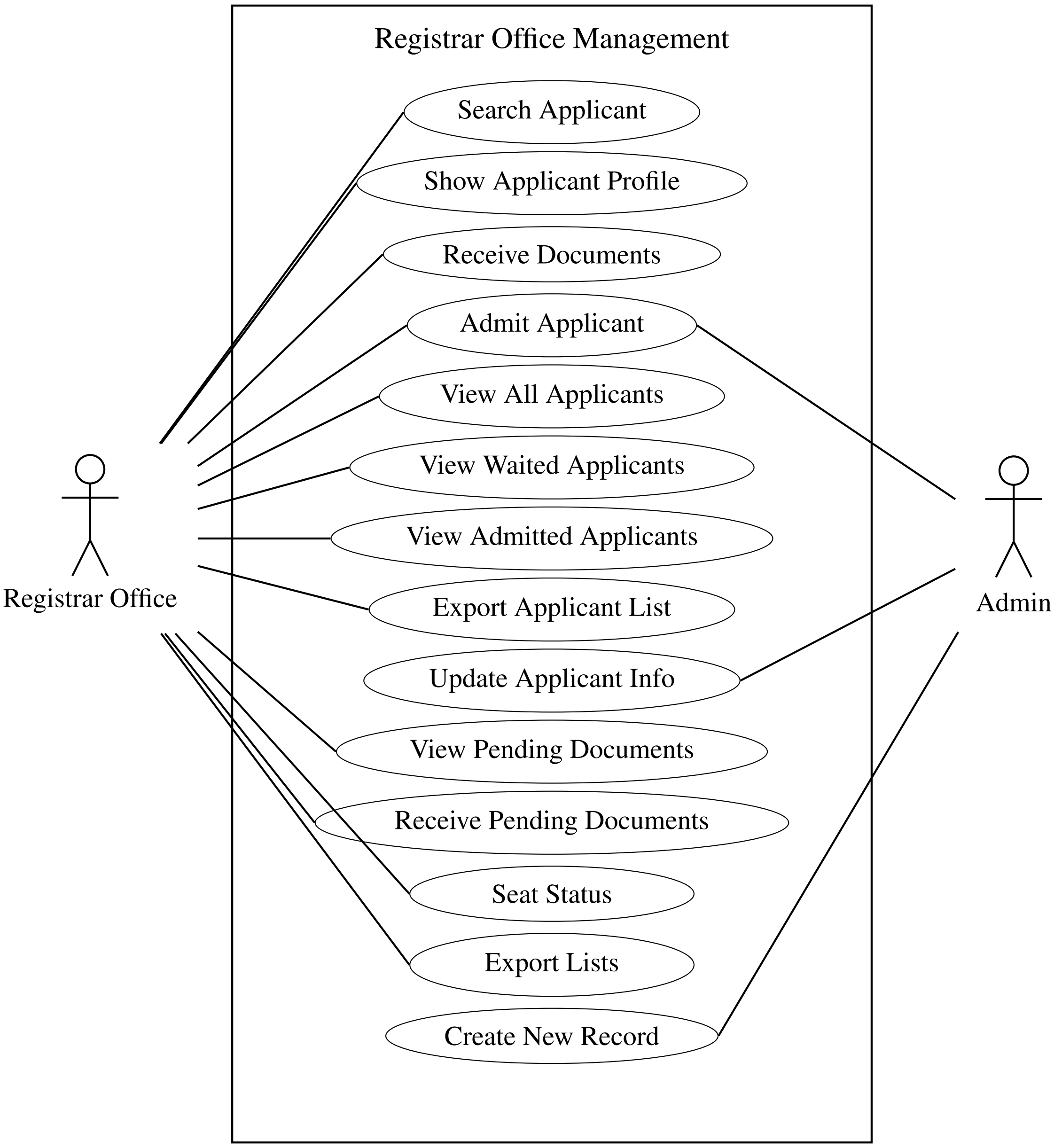


Figure 3.5: Use Case Diagram of Registrar Office Management

The Registrar Office Management use case diagram consists of two primary actors: Admin and Registrar Office. This module controls operations including searching applicants, viewing profiles, receiving documents, and monitoring seat status.This module also guarantees that all records relating to admission will be appropriately sorted out and easily available for review. Applicant's performance at various stages of the admission process will be effectively monitored. It will also allow the Registrar's Office to authenticate all submitted data according to the set standards before acceptance. Updates will be instantly available regarding application status so that any modifications

3.2.5.1 Search Applicant

The Search Applicant use case enables the Registrar Office to efficiently find details about an applicant by using either the unique ID or roll number. This enables the Registrar Office to easily find the details without having to manually scroll through massive amounts of data. This use case can only be accessed by authorized users of the Registrar Office once they log in successfully.

Table 3.22: Use Case Table: Search Applicant (UC-5.1)

Attributes	Details
Use Case ID	UC-5.1
Use Case Title	Search Applicant
Principal Actors	Registrar Office
Purpose	Find applicants' information quickly by using unique identification numbers.
Prerequisites	Login is required with the credentials of the Registrar Office.
Flow	1. Go to the Dashboard. 2. Access the search box or list of applicants. 3. Enter the applicant's ID or Roll No. 4. Click 'Enter' or Search button to retrieve results.
Exceptions	If the user does not have access or the entry does not exist in the system.
Frequency of Use	Used frequently by all administrative modules.

3.2.5.2 Show Applicant Profile

This particular use case will enable the Registrar Office to see the entire profile of any applicant of choice. Once the search for the applicant has been carried out, the user will then be able to click on the name of the applicant or the "View Profile" button to see all information about him/her in one particular page known as the profile. This use case is very useful especially when processing an applicant's application.Moreover, the profile layout is designed in a manner that organizes the information in categories such as personal information, academic information, and documents, making it easier to read. The system enables the Registrar's Office to evaluate all the information provided without the need to navigate from one webpage to another. The system provides a proper arrangement of all documents submitted in support of the application, making it easier to verify all documents. Additionally, all changes made to the information provided are updated instantly on the applicant's profile page.

Table 3.23: Use Case Table: Show Applicant Profile (UC-5.2)

Characteristics	Description
Use Case ID	UC-5.2
Use Case Name	Show Applicant Profile
Primary Actors	Registrar Office
Goal	To see the entire personal and academic profile of a particular applicant.
Precondition	The user is required to look up a certain applicant in the database.
Scenario	1. Look up a certain applicant. 2. Click on either the "View Profile" link or the name of the applicant. 3. The system presents the full profile page.
Exception	No authorization or no matching record.
Frequency of Use	High frequency use in the data and admission process.

3.2.5.3 Receive Documents

The Receive Documents use case allows the Registration Office to make an official record for the receipt of necessary documents. The user simply has to select the candidate from the database and receive the document using the "Receive Document" action. The application will update the record of the candidate indicating that he/she has received his/her documents. This feature provides a means of keeping track of admission documents.

Table 3.24: Use Case Table: Receive Documents (UC-5.3)

Characteristics	Description
Use Case ID	UC-5.3
Use Case Name	Receive Documents
Primary Actors	Registrar Office
Goal	To officially record and confirm that the documents have been submitted.
Precondition	Applicant must be located and chosen from the database.
Scenario	1. Locate a certain applicant. 2. Select him/her from the result list. 3. Click the "Receive Document" button to indicate his receipt.
Exception	Lack of compulsory document and invalid user role.
Frequency of Use	High frequency use at the document verification process.

3.2.5.4 Admit Applicant

The Admit Applicant use case enables the Registrar Office to formally admit the eligible applicants to the system. Once the applicant and eligibility of the selected applicant is confirmed, then the registrar will move on to admit the applicant. Through clicking the Admit button, the system will formally make the applicant an admitted individual. It is a vital part of the admission process and is commonly used when admitting applicants.

Table 3.25: Use Case Table: Admit Applicant (UC-5.4)

Characteristics	Description
Use Case ID	UC-5.4
Use Case Name	Admit Applicant
Primary Actors	Registrar Office
Goal	To finalize the process of admitting an applicant who is eligible.
Precondition	The applicant's details must be processed by the registrar.
Scenario	1. Find the particular applicant. 2. Ensure the eligibility of the applicant. 3. Press the 'Admit' button to admit the applicant in the system.
Frequency of Use	High frequency during admissions.

3.2.5.5 View All Applicant

The View All Applicants use case is used by the registrar's office to get information about all the registered applicants of the system. The user logs into the dashboard and uses this particular use case via the sidebar menu. The system provides all the details of applicants in a well-organized manner. It is very useful in checking the overall status of applications and carrying out admission process effectively.Moreover, the use case also enables the Registrar Office to sort applicants according to certain parameters such as admissions, completeness of submission, and other eligibility criteria. Such an approach gives a more organized tabular form of representation, making it easier to handle larger amounts of data. The application also enables search functionalities and pagination in order to find particular records faster. Moreover, each applicant record can be accessed separately whenever necessary. Additionally, updates in data are done in real time, ensuring consistency throughout the entire admission process since any update made in another office will be immediately visible. It also eliminates the need for additional record keeping in order to manage applicant data manually.

Table 3.26: Use Case Table: View All Applicant (UC-5.5)

Characteristics	Description
Use Case ID	UC-5.5
Use Case Name	View All Applicant
Primary Actors	Registrar Office
Goal	To gain a holistic view of all the registered applicants within the system.
Precondition	User must be logged into the dashboard.
Scenario	1. Log in to the dashboard. 2. Go to the sidebar menu. 3. Click on 'All Applicants'.
Exception	Unauthorized access.
Frequency of Use	High frequency during the entire admissions period.

3.2.5.6 View Waited Applicant

The use case of View Waited Applicant enables the Registrar Office to track the applications of candidates that have been put on hold for admissions. Once the system is logged in, one can click on the option through the dashboard sidebar. All waitlisted applicants will be displayed on a list for easy management by the registrar. This use case is common during the final stages of admission.In addition to this, this system will allow Registrar Office to view detailed information about each waitlisted candidate in terms of academics, criteria eligibility, and earlier verification status. It will help in making it clear whether the candidate belongs to the category of those who were accepted, denied, and waitlisted. There will be various filters through which applicants can be sorted according to their priority, merit, or availability of seats. It will also facilitate quicker access to the profile of each applicant for further consideration in case seats are available.This will ensure that all deserving candidates are not missed out during admissions. In addition to this, the feature related to waitlisting will help in making immediate updates regarding the status of each admission.

Table 3.27: Use Case Table: View Waited Applicant (UC-5.6)

Characteristics	Description
Use Case ID	UC-5.6
Use Case Name	View Waited Applicant
Primary Actors	Registrar Office
Goal	To monitor and control the list of applicants currently on the waiting list for admission.
Precondition	User should have logged in to the system using Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Go to the sidebar menu. 3. Click on 'Waited Applicant' to see the priority list.
Exception	Unauthorized login or session timeout.
Frequency of Use	High frequency when seat allocation and admission finalize.

3.2.5.7 View Pending Document Applicant

The View Pending Document Applicant use case enables the Registrar Office to see those applicants who still haven't finalized or verified the necessary documents. Once the user logs into the system using valid credentials, he/she is able to access the aforementioned functionality through the menu on the dashboard screen. In this use case, the system presents a list of applicants whose document is in pending state. This helps in locating those applicants whose application forms are still incomplete.The system also offers relevant data about individual applications, pointing out which specific documents are not available or need verification. Consequently, the Registrar Office can act specifically without having to analyze the whole application again. The system also ensures fast access to the profile of an individual applicant whenever additional checking is required. Finally, the list of documents to be added is updated in real time to indicate that the application has already been updated with new documents. Thus, the Registrar Office will have access to the latest data at all times. Applications with missing documents will be prioritized depending on how urgent or numerous these problems are. It will save considerable amount of time since incomplete applications will already be sorted.

Table 3.28: Use Case Table: View Pending Document Applicant (UC-5.10)

Characteristics	Description
Use Case ID	UC-5.10
Use Case Name	View Pending Document Applicant
Primary Actors	Registrar Office
Goal	Access and review the list of applicants who have not yet completed their document submission or verification.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the sidebar menu. 3. Click on 'Documents Pending' to view the specific applicant list.
Exception	Unauthorized access or session timeout.
Frequency of Use	High frequency during the document verification phase.

3.2.5.8 Export Applicant List

Export Applicant List is a use case that enables the Registrar's office to export data regarding applicants in a structured format of Excel sheets. Once logged into the system, the user can navigate the applicant list on the system's dashboard. There is an export feature that allows users to export either selected applicants' details or all applicant details within the system.

Table 3.29: Use Case Table: Export Applicant List (UC-5.8)

Characteristics	Description
Use Case ID	UC-5.8
Use Case Name	Export Applicant List
Primary Actors	Registrar Office
Goal	Exporting applicant list into an Excel spreadsheet format.
Precondition	User is logged in and has access to applicant records.
Scenario	1. Login to the dashboard. 2. Go to the Applicant table. 3. Choose your desired data and click the 'Export' button.
Exception	Unauthorized access or system error during export process.
Frequency of Use	High frequency

3.2.6 Applicant Profile Management

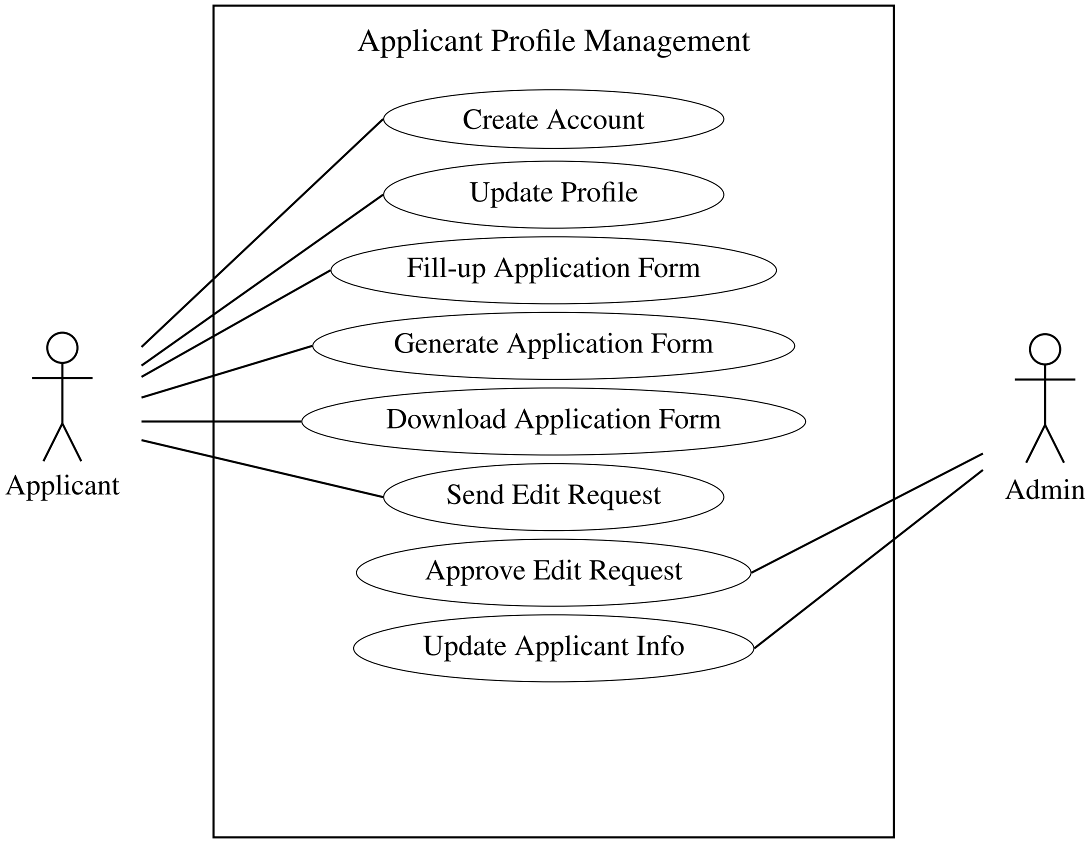


Figure 3.6: Use Case Diagram of Applicant Profile Management

The Applicant Profile Management module manages the student's interaction with the system, covering everything from registration to document retrieval.this module makes sure that the information of every applicant is accessible and updated by them whenever they want. The module serves as a platform where one can check the status of their application instantly. This module also gives users the ability to upload, view, and Applicant Profile Management module will handle all interactions of the student with the software right from the point of application till the moment of document acquisition. It is important to note that through this module, information about each applicant is made available to him or her at all times. The Applicant Profile Management module provides a platform where the applicant's application status can be checked immediately. The user through this module can manage the uploading and viewing of his or her personal and educational documents. Consistency of data throughout the entire application process will be ensured through this module. The applicant will also be able to update his or her profile data whenever needed.

3.2.6.1 Create Account

The “Create Account” use case is where a new user registers into the admission system. A first-time user needs to input their GST Applicant ID together with the necessary personal data. After inputting their information, the user submits the application and creates an account. After verification of the information, an account will be created by the system. This helps the applicants to log into the system for further admission-related activities.

Table 3.30: Use Case Table: Create Account (UC-6.1)

Characteristics	Description
Use Case ID	UC-6.1
Use Case Name	Create Account
Primary Actors	Applicant
Goal	Creation of new account in admission system.
Precondition	Must be first-time user.
Scenario	1. Enter GST Applicant ID details. 2. Enter all required information. 3. Press ”Register” button.
Exceptions	Invalid GST Applicant ID or Mismatched Data.

3.2.6.2 Update Profile

The Update Profile Use Case provides the ability for the user to update the information in order to keep it up-to-date and correct since the information entered by the user should be actual. As soon as the user logs in, he/she gets access to the main dashboard where one can find a possibility to move to the profile page. It is possible to change the information by entering data in the required fields. This use case allows validating data in order to make sure that it is valid and correct. Updating data prevents any possible errors that can occur during application and ensures that all the data that you enter is actually the latest one. In addition, a confirmation message can also be received as a response to an update request.The Update Profile Use Case allows users to have the opportunity to update data so as to have it up to date and accurate, as the data entered into the system is supposed to be actual. As soon as the user logs into the system, he or she gains access to the main dashboard, from which it will be possible to navigate to the profile page. Changing of data becomes possible when the user enters information in the designated fields. Validation of data becomes possible through the use case in order to confirm the accuracy of data being updated. Updating of data avoids any possible errors during the application process while ensuring that the data entered is current.

Table 3.31: Use Case Table: Update Profile (UC-6.2)

Characteristics	Description
Use Case ID	UC-6.2
Use Case Name	Update Profile
Primary Actors	Applicant
Goal	To allow applicant update and keep profile information updated.
Precondition	User must log-in into account.
Scenario	1. Go to Applicant Dashboard page. 2. Press Profile Icon. 3. Select ”Update Profile” from menu. 4. Enter all required information. 5. Press ”Update” button to update changes.
Exceptions	Connection Error and/or Expired Session.

3.2.6.3 Fill-up Application Form

The use case of Fill-up Application Form enables the applicant to fill in and submit the application form within the system. Upon signing in to the dashboard, the applicant gains access to the form filling part of the process. Multiple stages are involved in order to ensure that all necessary details are entered properly by the user. Upon completion of all stages, the form is submitted.

Table 3.32: Use Case Table: Fill-up Application Form (UC-6.3)

Attributes	Attributes’ Description
Use Case ID	UC-6.3
Name of Use Case	Fill-up Application Form
Primary Actor	Applicant
Goal	Successfully fill and submit the admission application form.
Precondition	Applicant should log-in on the dashboard.
Scenario Steps	1. Access Applicant Dashboard. 2. Click ‘Fill-up Application Form’. 3. Provide information in all five sequential steps. 4. Submit the form.
Exceptions	Invalid role or session timed out.

3.2.6.4 Generate Application Form

Use Case Description of "Generate Application Form" Generate Application Form use case enables applicants to download an electronic copy of their submitted application form. After successfully performing all necessary tasks in the application process, applicants can utilize this functionality on their dashboard page. The system analyzes all submitted information and creates the final application form. This form is then made available for downloading purposes by the users.

Table 3.33: Use Case Table: Generate Application Form (UC-6.4)

Attributes	Attributes' Description
Use Case ID	UC-6.4
Name of Use Case	Generate Application Form
Primary Actor	Applicant
Goal	To create the downloadable form of the submitted application form.
Precondition	Applicant should successfully fill up all five steps of the application form.
Scenario Steps	1. Visit the Applicant Dashboard. 2. Click 'Generate Application Form' button.
Exceptions	Incomplete form steps or invalid role.

3.2.6.5 Download Application Form

The Download Application Form use case enables the applicant to download the application form for admissions in pdf format. After logging into the system, the user will find this use case from the dashboard of the application. The system will produce the final application form and allow downloading of the same. This use case makes sure that the applicant will have easy access to his/her application form for further references.Download Application Form is the use case that allows the applicant to download the application form for admissions in a PDF format. After signing on to the system, the user can access this use case via the dashboard in the application. The system creates the application form and also downloads the document for the applicant. This use case guarantees that the applicant gets an easy access to his/her application form for further uses. The use case also guarantees that the applicant gets access to the generated document containing all the information that has been updated by the system after verifying it. The system automatically integrates the user details in a standardized manner. Preview facility is available to applicants to verify that the form downloaded has been created with accurate information. The download facility ensures that downloading process does not put much burden on the system and remains fast for applicants. Access control security measures are adopted to prevent access by unauthorized users. The system generates the form in the standard format of PDF which is printable. Each generated document gets the timestamping feature for documentation purposes. In case of incomplete information, the system notifies the user to update his/her information.

Table 3.34: Use Case Table: Download Application Form (UC-6.5)

Attributes	Attributes' Description
Use Case ID	UC-6.5
Name of Use Case	Download Application Form
Primary Actor	Applicant
Goal	To download the admission application form in PDF format.
Scenario Steps	1. Visit the Applicant Dashboard. 2. Click 'Download Application Form' button.
Frequency of Use	High

3.2.6.6 User Management

The User Management activity diagram represents the process flow of the system's administration and user functionalities. Admin and users need to sign in using their respective logins and passwords. After authentication, the system branches out depending on the user roles and operations to be performed. In the case of administrative users, the system offers various options for managing their tasks including verifying user credentials, assigning roles, and tracking system activities. The administrator can assess applications received, authenticate the validity of the information provided by the users, and either accept or deny the request based on predefined rules.

In the case of regular users, the system enables users to have a personal interface through which they can manipulate their accounts, apply for services, and monitor the status of the applications made. Users are also permitted to modify their personal information and add necessary files whenever required. The system performs proper validations during the process to ensure consistency and correctness of the data entered. In the event the details are not accurate, the user is asked to correct his/her information before progressing any further. Notifications are sent out to alert the users of any notifications, acceptance, or pending issues.

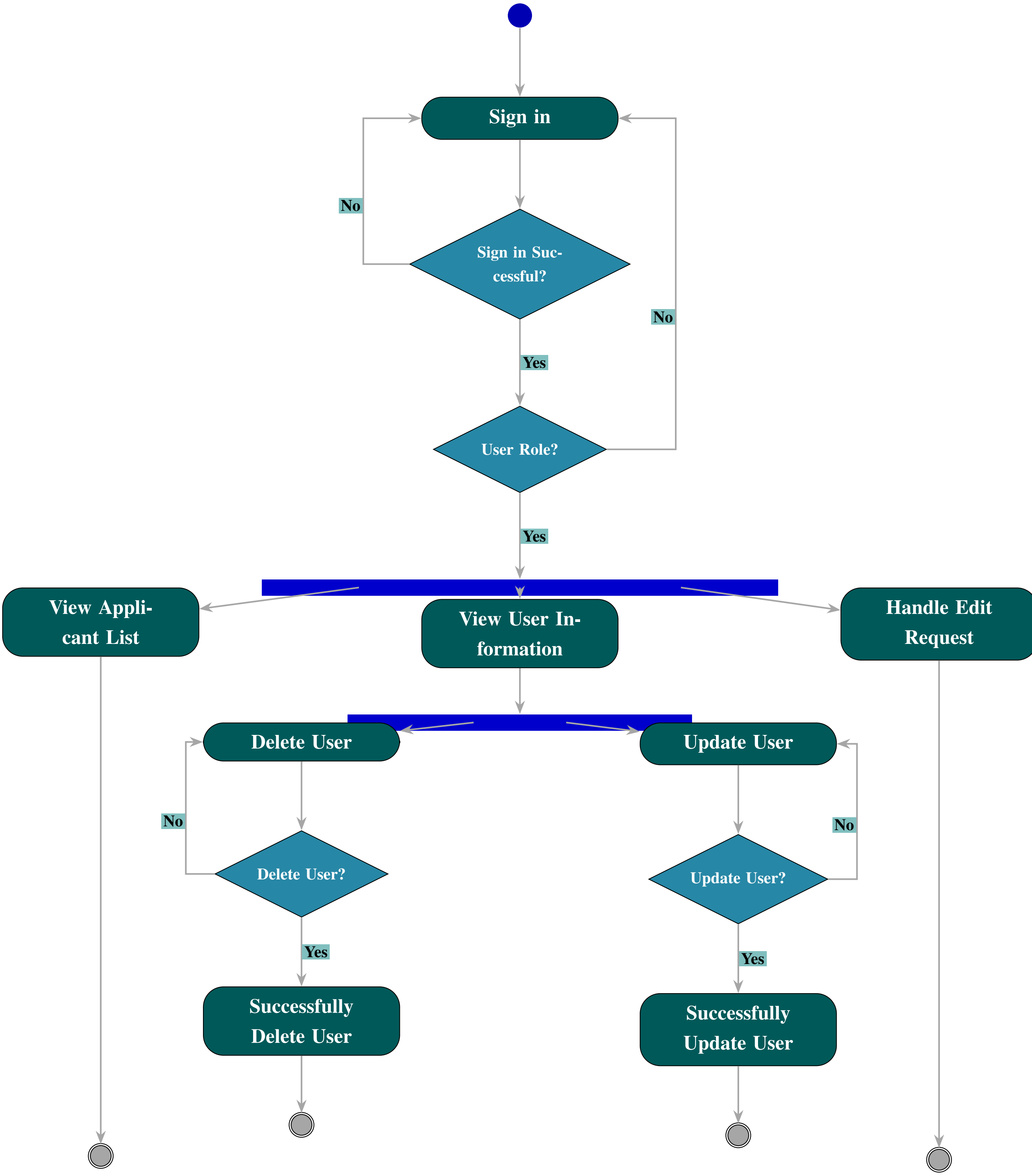


Figure 3.7: Activity Diagram of User Management System

3.2.6.7 Dean Office Management

The Dean Office Management activity diagram illustrates the workflow for faculty-level admission administration. Once the Dean or Registrar signs in successfully, the system allows for diverse operations including applicant searching, document verification, and seat status monitoring.

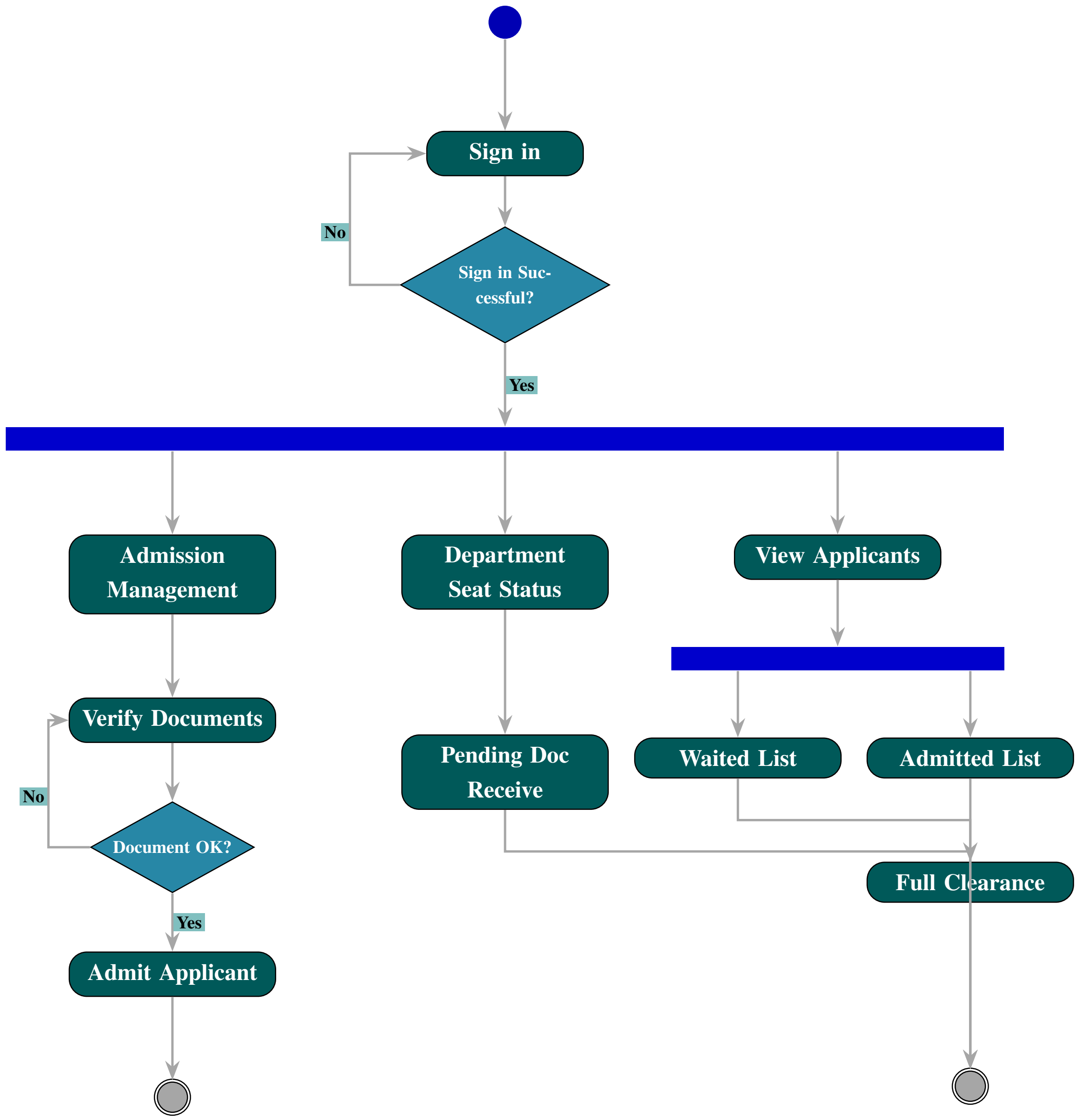


Figure 3.8: Activity Diagram of Dean Office Management (UC-5.0)

3.2.6.8 Hall Office Management

The Hall Office Management activity diagram illustrates the operational workflow for residential hall admission tasks. It covers the process from secure authentication to applicant search, document receipt, and final admission, as well as administrative tasks like exporting applicant lists and downloading official records.

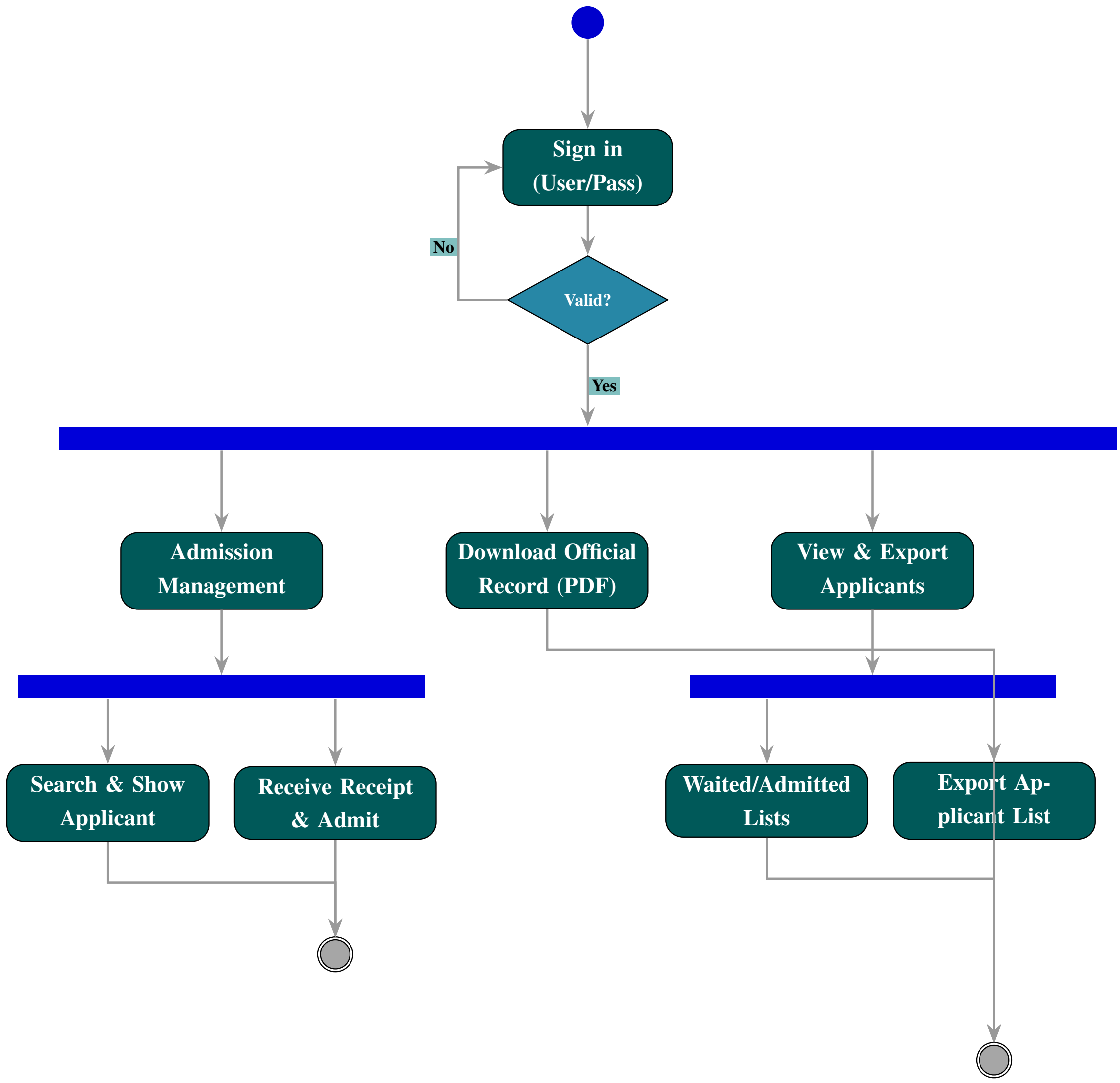


Figure 3.9: Activity Diagram of Hall Office Management (UC-6.0)

3.2.6.9 Medical Office Management

The Medical Office Management flow chart shows the process involved in handling the medical record and admissions for the applicants. Upon signing in successfully, the medical officer is allowed to look for applicants, update their medical records, produce medical reports, and handle the admitted applicants and those on waitlist. The system ensures that all medical updates are validated before submission to avoid any errors in data management. Moreover, the officer will also have access to check all pending documents to ensure that all applicants meet the necessary medical requirements. Reports produced by the system can be saved either in print or digital format.

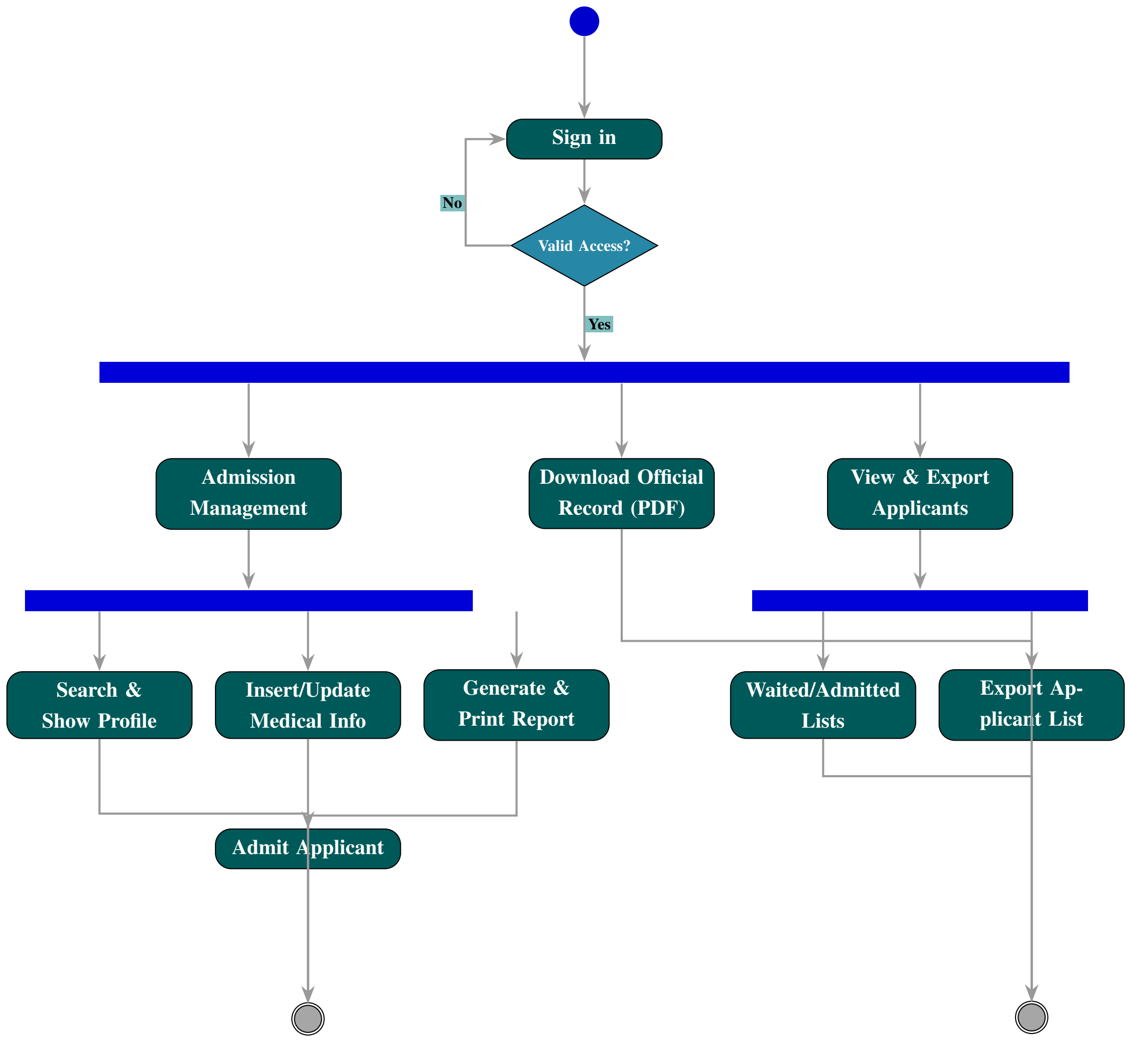


Figure 3.10: Activity Diagram of Medical Office Management (UC-4.0)

3.2.6.10 Registrar Office Management

The Registrar Office Management activity diagram depicts the general administration tasks that the registrar will be involved in. These include authentication, management of applicants (search, submission of documents, and admitting of the applicants) and advanced reporting functionalities such as list exportation, creation of customized records for official purposes. Only authorized personnel will be able to access any administrative function in the system after successful authentication through login. The system is designed to efficiently handle huge amounts of applicants' information with minimal efforts. The registrar will also have the option of sorting and grouping applicants depending on their admission level such as waitlisted or admitted or other pending cases. The system ensures data consistency through validation of all information prior to being saved in the database.

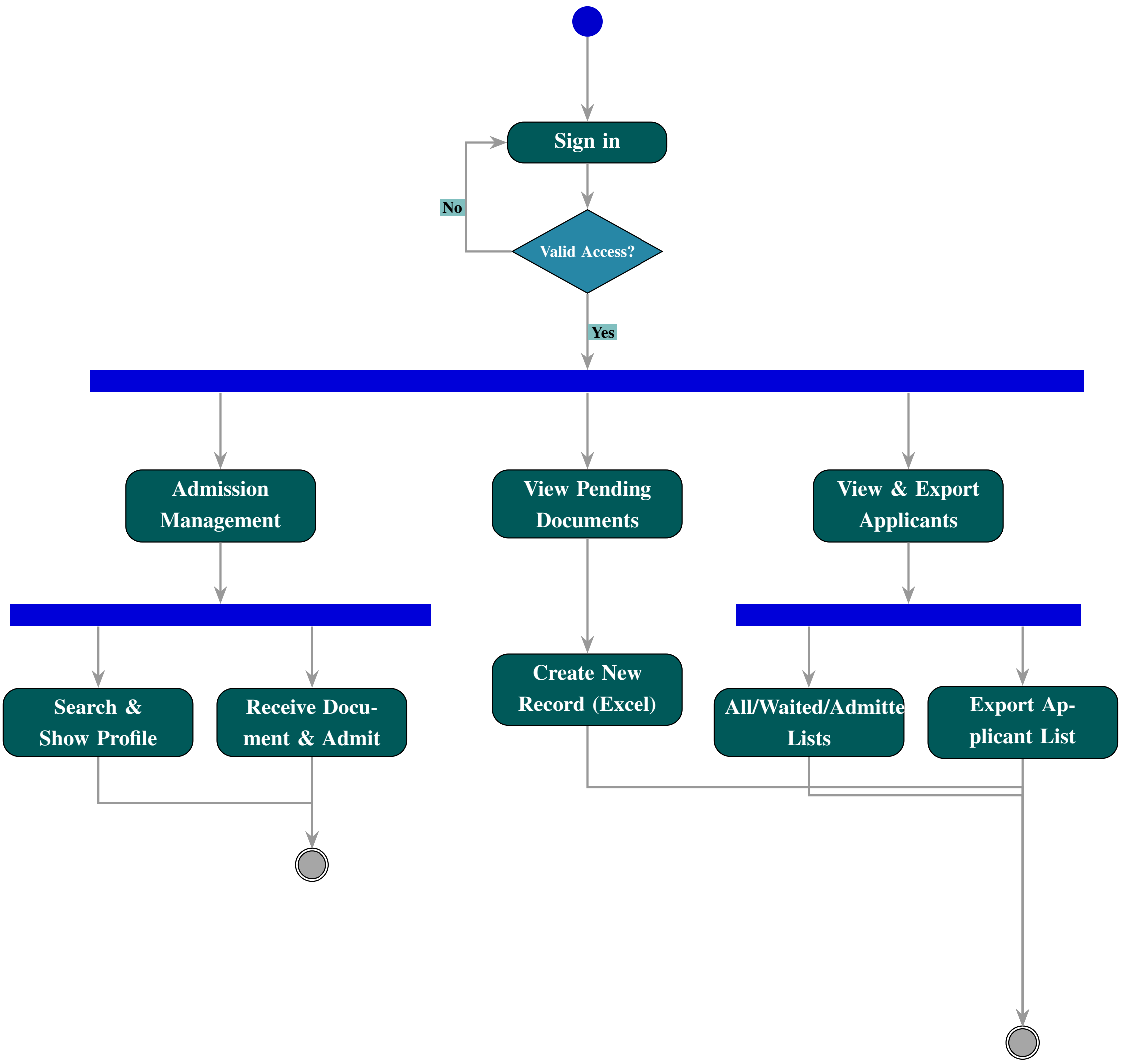


Figure 3.11: Activity Diagram of Registrar Office Management (UC-7.0)

3.2.6.11 Applicant Profile Management

The Applicant Profile Management activity diagram illustrates the interactive workflow for students using the admission system. It encompasses the secure login process using GST credentials, profile updates, the multi-step application form filling process, and the final generation and download of the application form, including the ability to send edit requests post-submission. The system ensures that each step of the application process is completed in a structured and validated manner to reduce errors and incomplete submissions. Applicants can update their personal information whenever necessary before final submission, maintaining data accuracy. During the form-filling stage, the system guides users through all required sections to ensure completeness. After submission, applicants can still request modifications through an edit request mechanism, which is reviewed by the authority. The generated application form can be downloaded for future reference and official use. Overall, this module provides a user-friendly and controlled workflow that enhances efficiency and reliability in the admission proces

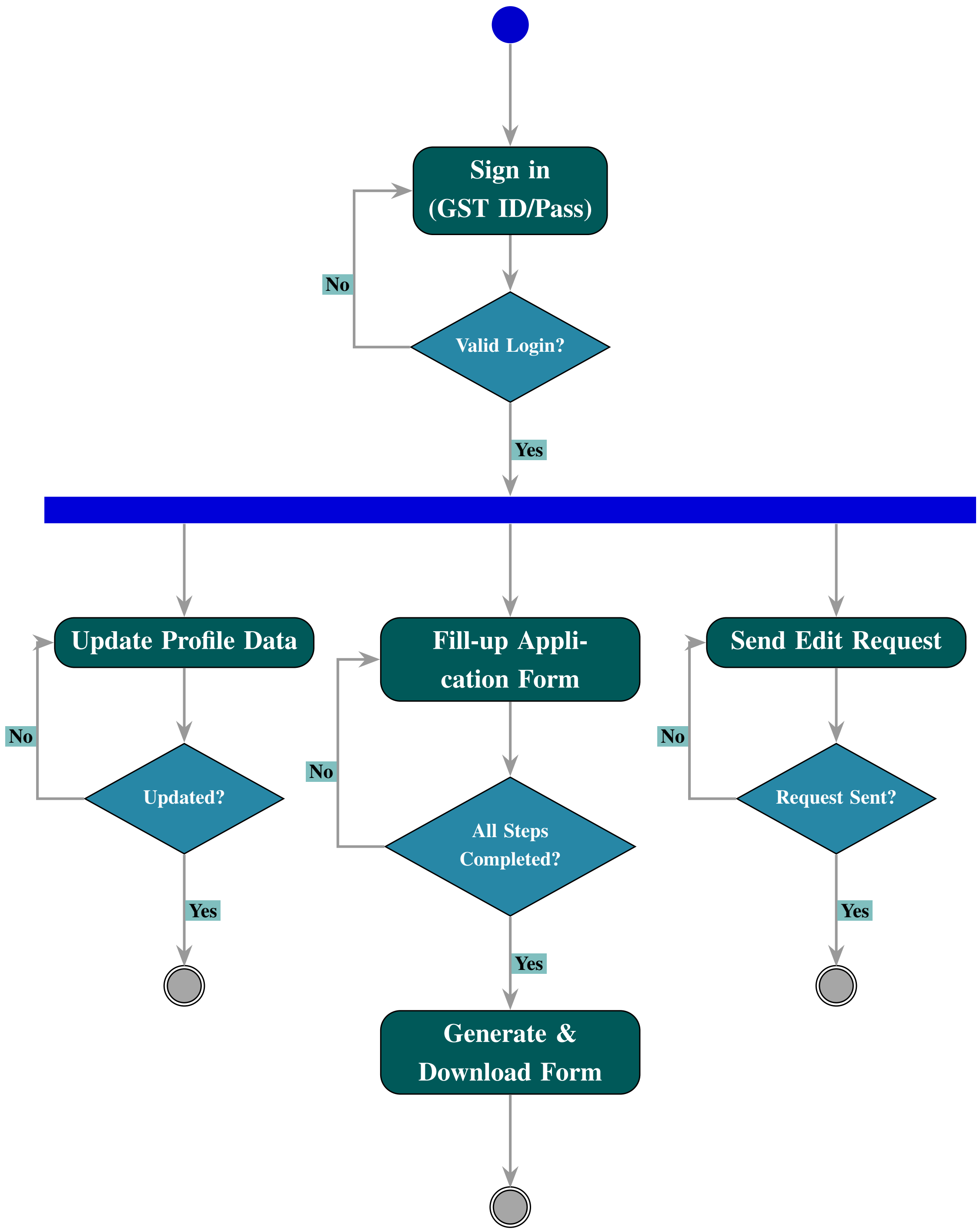


Figure 3.12: Activity Diagram of Applicant Profile Management (UC-2.0)

3.2.7 System Database Schema

The overall database architecture and entity relationships of the JUST Undergraduate Admission System are presented in the following schema diagram. This diagram illustrates the interconnections between user authentication, academic records, and administrative clearance modules.

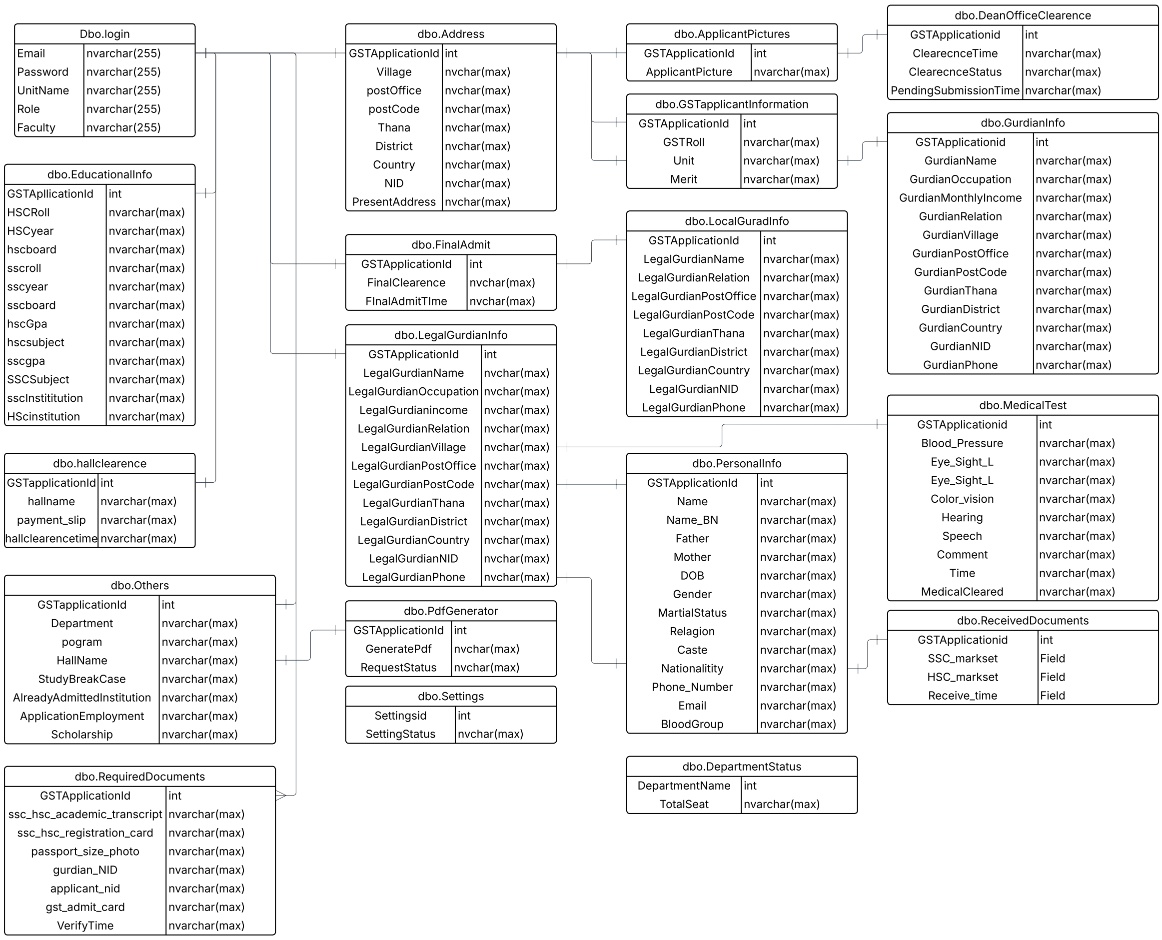


Figure 3.13: Full Database Schema Diagram of the System

Chapter 4

Implementation Details

4.1 Chapter Overview

In this chapter, the discussion will be about the technical design of the various functional modules starting with the Authentication Module. This program creates a secured environment where access controls are implemented based on Role-Based Access Control (RBAC). The Students Applicants Module is designed for the creation of profiles, handling multi-form applications, and automatic PDF file creation. Lastly, the Administration Module will act as the main module which facilitates all the data-related functions as well as monitoring functions in the system.
Other areas included in the development of this implementation will be special workflow designs for Dean, Registrar, Hall, and Medical modules. Through these modules, applicants can be verified, and all the required data can be validated prior to making the admission official. At last, we will discuss in this chapter the process for System Integration, focusing on integration with cloud storage and other third-party systems.

4.2 Introduction

This chapter discusses the detailed implementation aspects of the Digital Admission Procedure Management for Post-GST Candidates (JUST-DAPM-PGC). It presents the actual user interfaces developed, focusing on the workflows, data handling, and specific functionalities of different user roles, starting with the authentication process and subsequent role-specific modules.

4.3 Authentication

4.3.1 Sign in Page and Process

Sign in Page and Process

The system needs to make sure that users are really who they say they are. This is a security check to keep everything. It makes sure that authorized people can use the features of the sign in page and process.

The system has a sign in process that all users need to follow. This includes students, administrators and office staff who all have to sign in.

The sign in process is very easy to use. It works well for everyone. The sign in page and process are available in two languages: English and Bengali.

The sign in page looks modern and clean which makes it easy to use the sign in page and process.

Here's how to sign in to the sign in page and process:

- Enter your details: You need to enter your ID and password on the sign in page. Students can use their email or GST Admission Roll on the sign in page. The initial password for students is their HSC Roll, which they use on the sign in page.
- See your password: There is a button that lets you view what you type on the sign in page. This helps you check and avoid mistakes on the sign in page.
- Login: You click the Login button on the sign in page. This starts the verification process for the sign in page and process.
- Go to your page: If you sign in correctly to the sign in page the system takes you to your page. This could be the Student page, the Dean page or the Registrar page, which you can access after signing in to the sign in page and process.
- Error: If you sign in incorrectly to the sign in page the system does not let you in. It says that the username or password is incorrect for the sign in page and process.

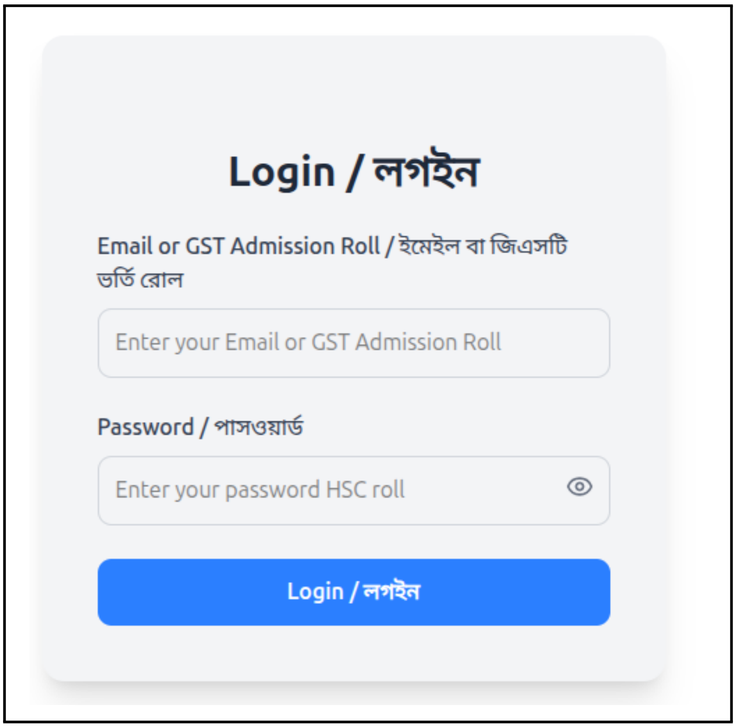


Figure 4.1: System Authentication (Login Page)

4.4 Student Module Implementation

When students log in successfully, they are directed to their application portal. The main task for a student is to complete the admission form by providing accurate personal and academic information.

4.4.1 Final Admission Form

The admission form is designed to collect detailed information about the student's home and identity. Each field is validated before submission to ensure the accuracy of the data.

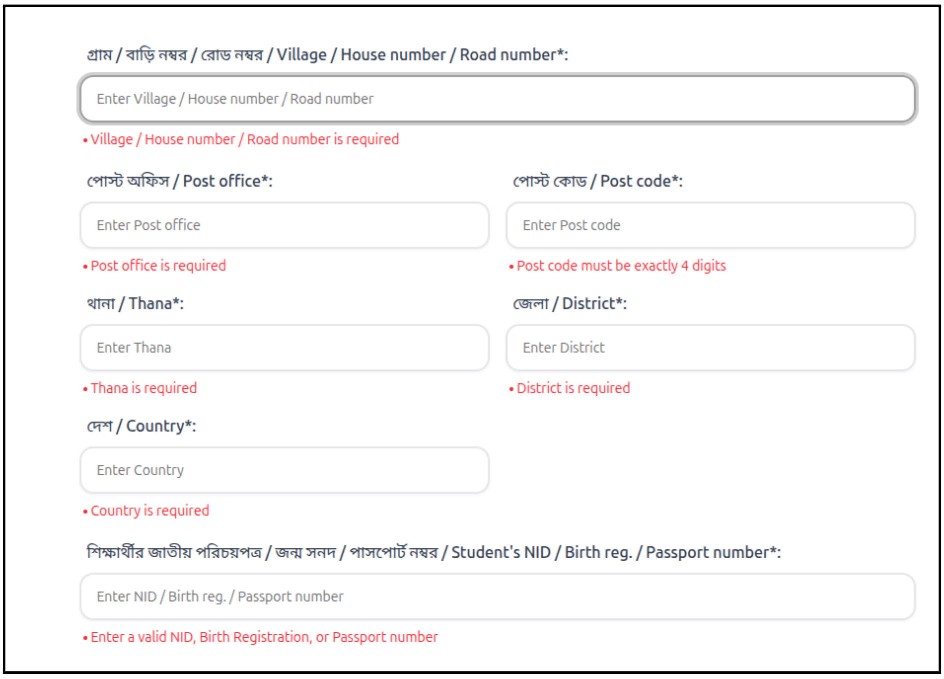


Figure 4.2: Student Admission Form with Validation

As shown in Figure 4.2, the form contains the following sections:

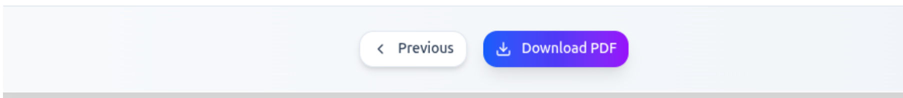
- Address Details: Students must provide their *Village/House number*, *Post Office*, *Thana*, *District*, and *Country*. Mandatory fields are marked with an asterisk (*).
- Postal Code Validation: The Post Code field requires exactly 4 digits, preventing incorrect entries.
- Identification Information: Students can provide either their NID, Birth Registration, or Passport number.
- Real-time Error Handling: The system immediately notifies the student of errors. If a field is empty or contains invalid data, a red error message appears to guide correction.

- Bilingual Interface: All labels are shown in both Bengali and English, ensuring all students can understand and complete the form accurately.

This structured approach ensures the database receives validated and formatted information, which simplifies verification by the Dean and Registrar offices. The admission form is the key component of the Student Module, which is essential for a smooth admission process.

4.4.2 Application Summary and PDF Download

After completing the necessary steps of the admission form, the student reaches the final stage where they can review their application summary and export the data for official use.



.png

Figure 4.3: Final Application Submission and Download Interface

The features of this section, as shown in Figure 4.3, are described below:

- Review Mechanism: The *Previous* button allows students to navigate back and double-check their provided information before final submission. This ensures that no incorrect data is sent to the university database.

- Instant PDF Generation: The system is integrated with a PDF generation library that converts the student's input fields into a formal document format. By clicking the Download PDF button, the user can instantly save a copy of their admission form.

- Visual Feedback: The button utilizes a distinct gradient color and a download icon to provide clear visual feedback, indicating that this is the primary action of the page.

- Official Requirement: This downloaded PDF acts as an acknowledgment slip, which contains a unique identifier for the student. It is mandatory for students to bring a printed copy of this document during the physical verification at the Dean's office.

4.4.3 Submission Confirmation and Complaint Management

When a student submits their application form they see a confirmation message. This message tells them that their information is safe in our database.

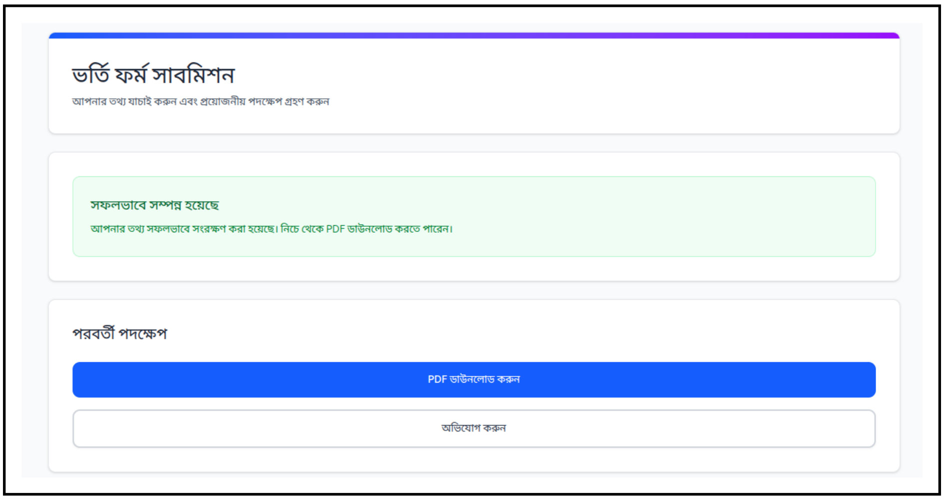


Figure 4.4: Form Submission Confirmation and Complaint Interface

The confirmation page has some features:

- Success Notification: A green alert box pops up. Says their information is saved. This gives students peace of mind.
- PDF. Download: Students can save their application form as a PDF. This way they can look at it again later. This feature is part of our report generation system.
- Complaint Management: If students find an error in their data or have issues they can report it away.
- Action Buttons: There are buttons like “Download PDF” and “Report a Problem”. These buttons help students know what to do.

This feature makes our system more reliable. It also helps students away and makes sure they have a good experience with the Digital Admission Procedure Management for Post-GST Candidates. The complaint management helps students report any issues with their application form. The system provides an experience, for students.

4.5 Admin Module: Data and Resource Management

The Admin module serves as the primary gateway for preparing the system for a new admission session.

4.5.1 Bulk Data Import and External Storage Integration

The administrator can populate the database by uploading Excel or SQL files containing GST admission results. To ensure data safety, a Google Drive integration is also provided.

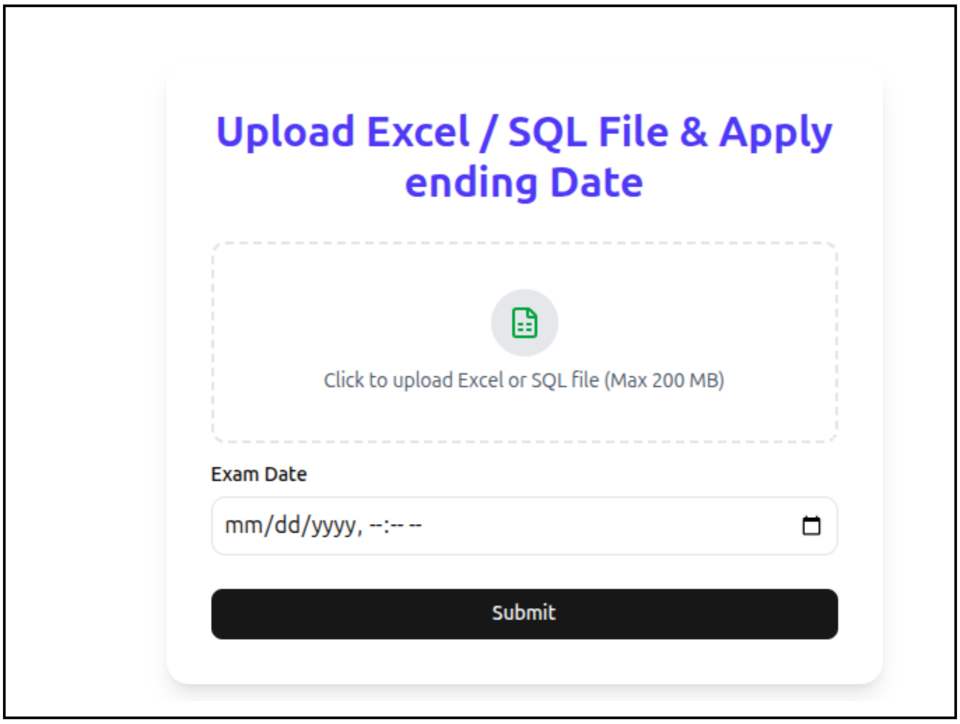


Figure 4.5: Excel/SQL File Upload and Application Deadline Interface

- Data Ingestion: The system parses the uploaded file and updates the applicant records. The interface also allows setting an "Exam Date" as a deadline.
- Cloud Backup: Using the interface shown in *uploadDrive.png* (if applicable), the admin can sync local data with Google Drive for disaster recovery.

4.5.2 Examination Announcement

The admin defines the specific admission unit and the corresponding examination date to notify the applicants.

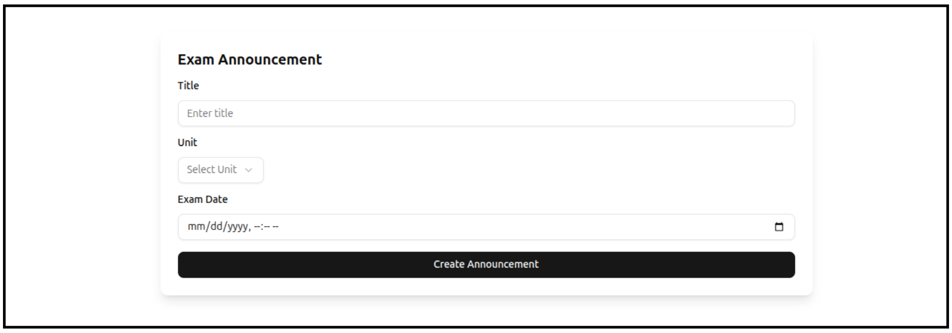


Figure 4.6: Exam Announcement Configuration Interface

4.6 Admin Module: User Control and Security

The people in charge of the university can manage security through a panel that controls who gets to do what.

4.6.1 Role-Based Credential Generation

The Admin can make accounts for the Deans and the Registrars and the Medical Officers using a special form just for that.

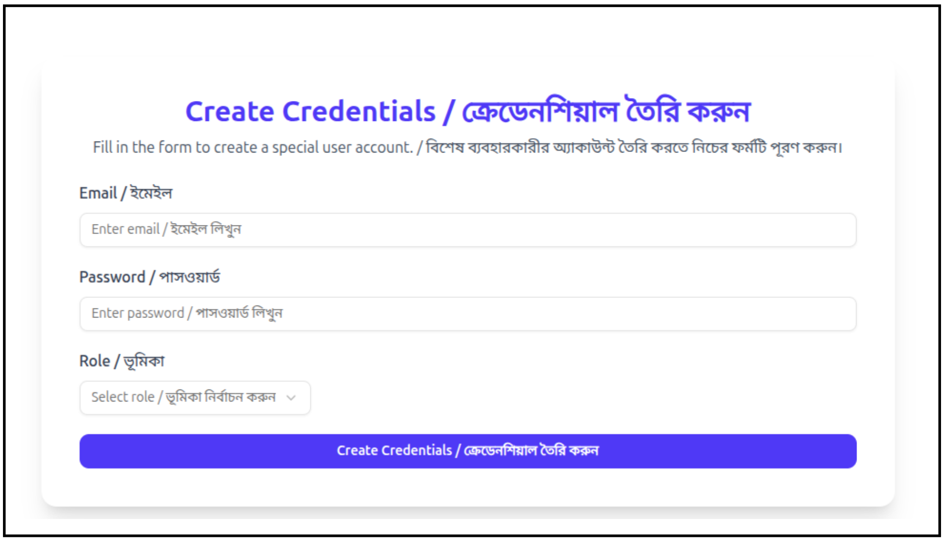


Figure 4.7: User Credential Creation Interface

The Admin can use the Admin Module to manage the User Control and Security. The Admin is in charge of the User Control and Security.

4.7 Admin Module: Complaint Resolution System

The school wants to help students with their problems. So the Admin panel has a system that tracks complaints in time. This system is for complaints that students send through the student portal.

4.7.1 Reviewing Pending Complaints

All the problems that students have are listed in a list. This helps the administrators deal with them one, by one.

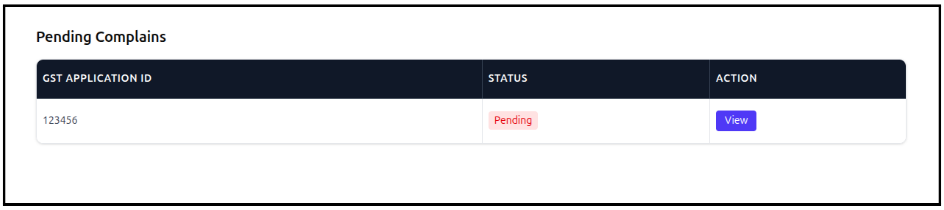


Figure 4.8: Pending Complaint Tracking Interface

When a complaint comes in the Complaint Resolution System does a things:

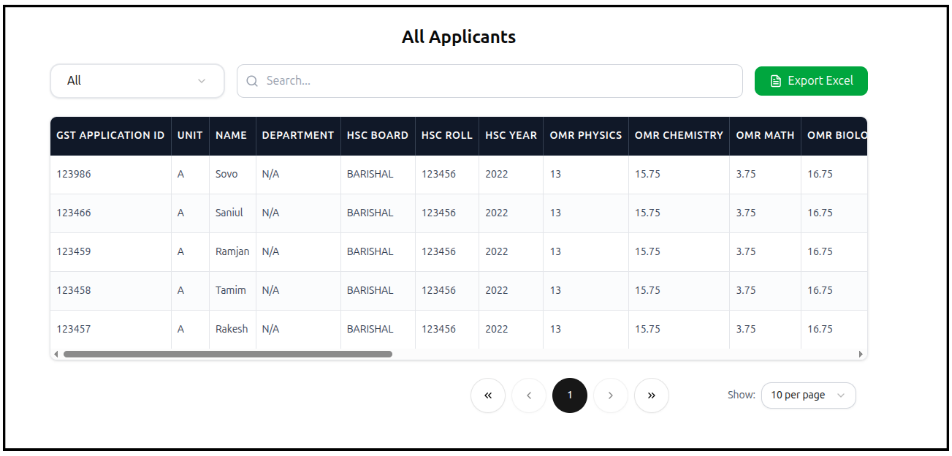
- The system matches the complaint to an Application ID: This is so the administrators can check the students records.
- The administrators can click the "View" button: To read the complaint. Then they can take the steps to fix the problem. The Complaint Resolution System is used to handle complaints. The Complaint Resolution System helps the administrators to deal with student complaints.

4.8 Dean Module: Applicant Verification

The Deans main job is to look at the list of applicants for each department and check if they are eligible to study.

4.8.1 Departmental Applicant Queue

The portal shows a list of students who are assigned to departments under the faculty. This list is updated in time.

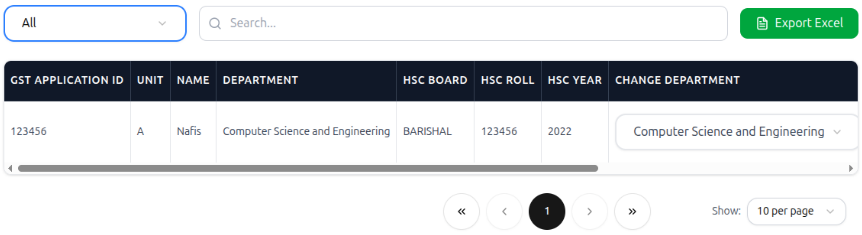


GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	OMR PHYSICS	OMR CHEMISTRY	OMR MATH	OMR BIOLO
123986	A	Sovo	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123466	A	Saniul	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123459	A	Ramjan	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123458	A	Tamim	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123457	A	Rakesh	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75

Figure 4.9: Dean's Portal: Integrated Applicant List

4.8.2 Detailed Information Table

To help the Dean verify the applicants properly there is a detailed table that shows information about each student.



GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	CHANGE DEPARTMENT
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Computer Science and Engineering

Figure 4.10: Dean's Detailed Data Verification Table

The Dean uses this table to check the GST scores and the requirements of each department. This helps the Dean decide if an applicant can move to the stage, which is approval. The Dean checks the GST scores and department requirements. The applicant list is checked by the Dean.

4.9 Dean Module: Student Status Management

The Dean Module is for managing the status of students. The Dean has the power to approve a student for a department or cancel their application if there are problems with their records.

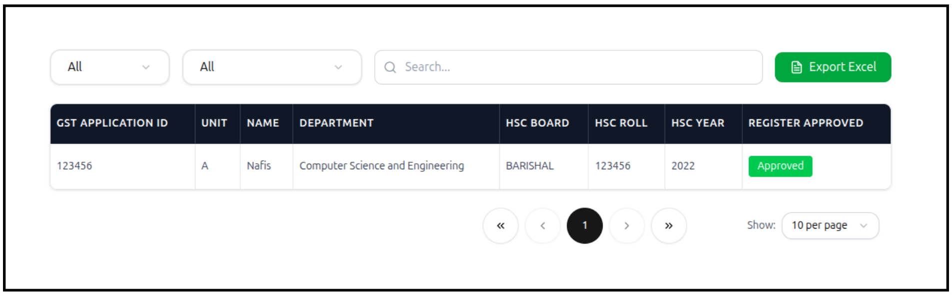


Figure 4.11: Dean's Portal: Admission Approval and Cancellation Interface

The Dean can do these things:

- Approve Admission: If a student's grades and documents are okay, the Dean can click the "Approve" button to send the student to the next step.
- Cancel Admission: If a student's information is fake or they do not meet the requirements of the department, the Dean can cancel their admission status.

The Dean Module is part of the Student Status Management system. The Dean uses the Deans Portal to make decisions about student admissions. The Dean's decisions are final and affect the students' status in the system. The Dean can cancel a student's admission at any time.

4.10 Registrar Module: Centralized Data Validation

The Registrar department possesses the responsibility for the final verification of applicant data before confirming admission.

4.10.1 Master Applicant Queue

The Registrar tracks the global applicant queue to ensure the smooth execution of the admission cycle within all the faculties.

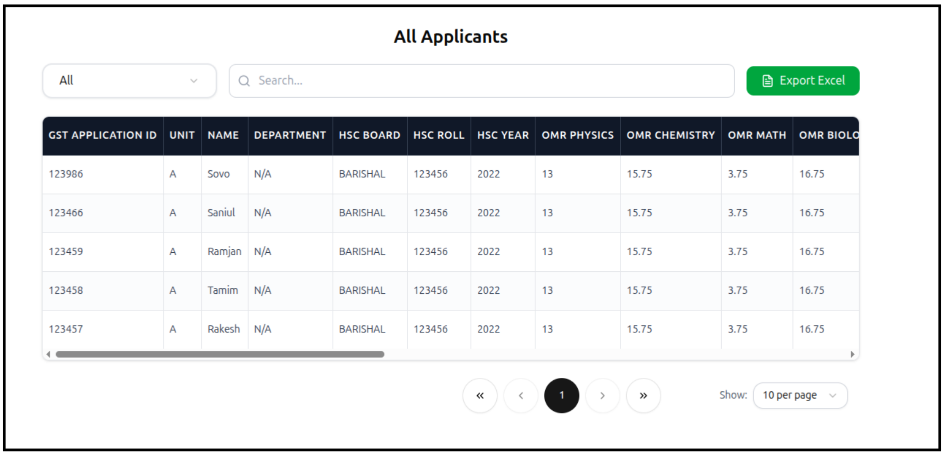


Figure 4.12: Registrar's Master Applicant Data Screen

4.10.2 Detailed Verification Matrix

In this case, the Registrar makes use of a detailed database to verify whether the phonological submission of documents is supported by their digital counterpart.

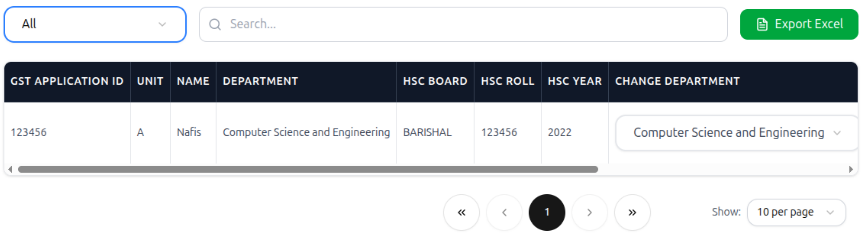


Figure 4.13: Registrar's Detailed Verification Table

Thus, the dual verification helps to ensure that only applicants who have valid academic qualifications and fee payments proceed to the next step.

4.11 Registrar Module: Data Archiving and Redundancy

Being the main keeper of records at the university, the Registrar module is concerned with data archiving and redundancy.

4.11.1 Automatic Back Up to Cloud Services

To reduce the risks of losing any information and have access to the data anytime, the system uses a direct uploading process to the university's official Google Drive.

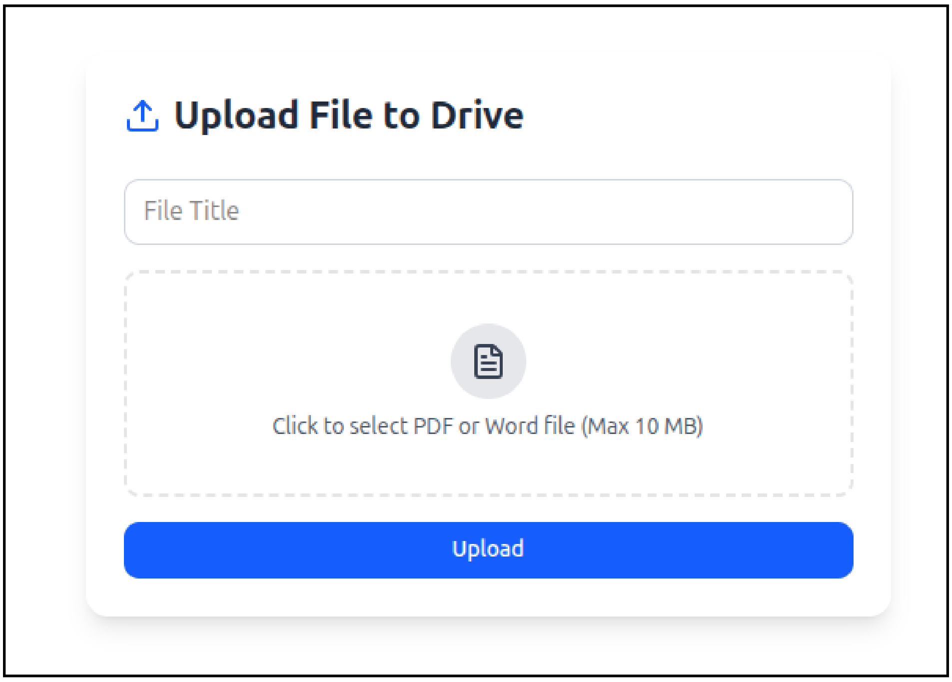


Figure 4.14: Registrar's Archive System (Google Drive Sync)

Using this interface, the Registrar will be able to back up all data from one session.

4.12 Registrar Module: Final Authorization

The Registrar performs the final audit. This interface allows the Registrar to finalize the admission or reject it if any legal or official requirement is missing.

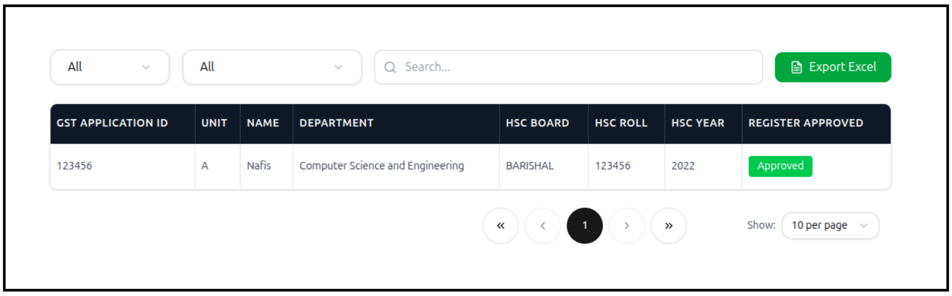


Figure 4.15: Registrar's Final Admission Authorization Dashboard

Through this dashboard, the Registrar ensures:

- Final Confirmation: Authorized students are officially enrolled in the university database.
- Admission Rejection: The Registrar can cancel or revert an admission if the payment or official verification fails at the last moment.

4.13 Hall Registrar: Residential Clearance

The Hall Registrar manages the residential status. Only students with an "Approved" hall status are allowed to reside in the university halls.

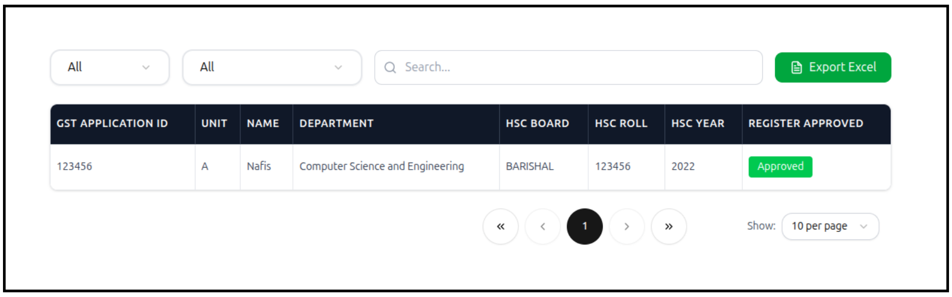


Figure 4.16: Hall Allocation Approval and Cancellation View

- Seat Approval: Assigning a hall name and room to the student.
- Allocation Cancellation: Removing a student from the hall list if they choose to stay off-campus or provide incorrect information.

4.14 Hall Registrar Module: Resident Data Verification

To ensure accurate record-keeping, the Hall Registrar utilizes a specialized queue to manage applicants who have applied for residential facilities.

4.14.1 Hall Applicant Queue

The interface displays a list of students who have cleared the academic and administrative stages and are now waiting for residential allocation.

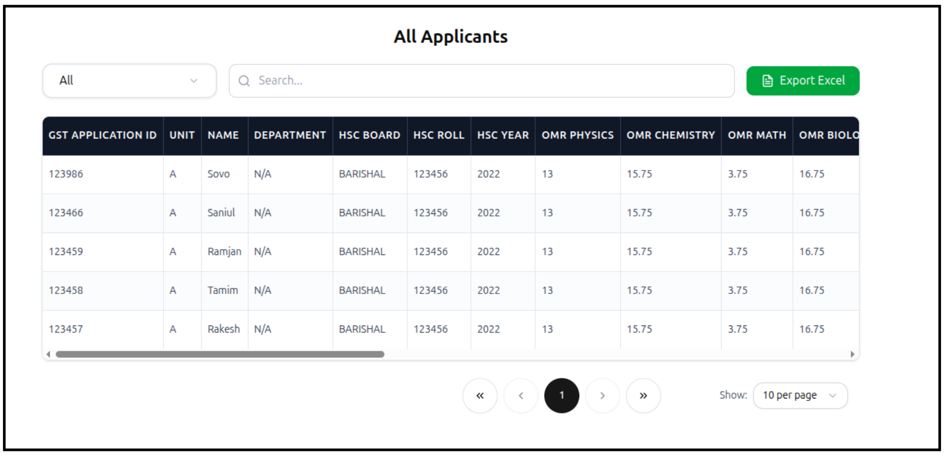


Figure 4.17: Hall Resident Applicant Queue

4.14.2 Detailed Resident Attributes

The Hall Registrar can view specific student data, such as permanent address and gender, which are crucial factors in hall assignment.

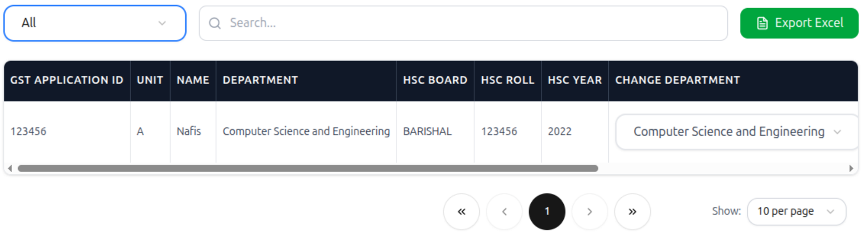


Figure 4.18: Detailed Resident Information Matrix

4.15 Hall Registrar Module: Redundant Data

Given the confidential nature of residential data, the Hall Registrar module will come with a special synchronization system to safeguard the information about allocations.

4.15.1 Cloud Synchronization of Hall Data

The Hall Registrar is capable of synchronizing the final residential allocation lists in the cloud storage for use by hall offices in allocating rooms to students.

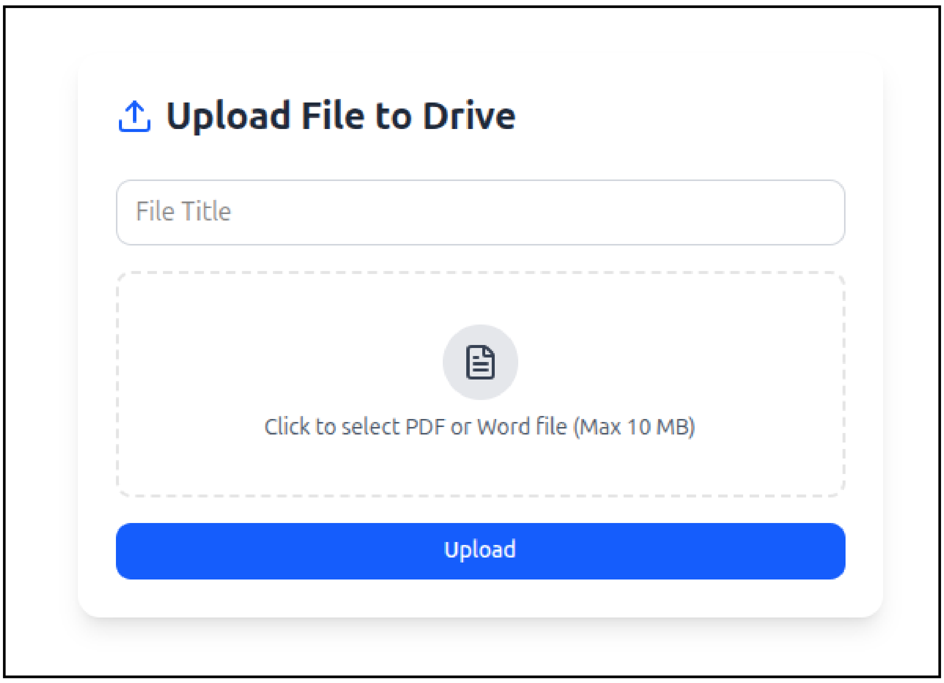


Figure 4.19: Hall Office Data Synchronization and Backup

4.16 Hall Registrar: Residential Clearance

The Hall Registrar manages the residential status. Only students with an "Approved" hall status are allowed to reside in the university halls.

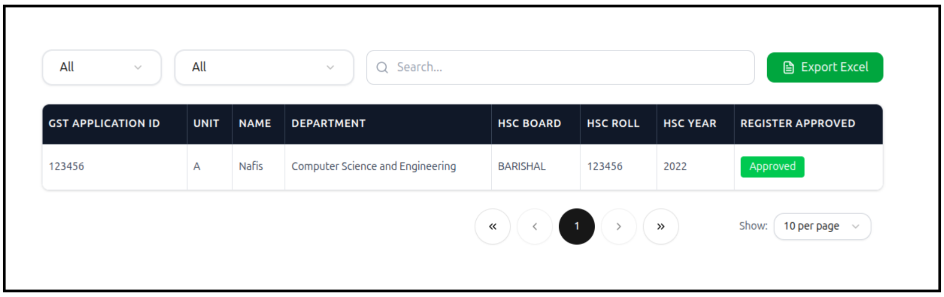


Figure 4.20: Hall Allocation Approval and Cancellation View

- Seat Approval: Assigning a hall name and room to the student.
- Allocation Cancellation: Removing a student from the hall list if they choose to stay off-campus or provide incorrect information.

4.17 Medical Officer Module: Applicant Health Data

This module enables the Medical Officer to view the list of applicants that are yet to undergo medical examinations or medical record validation.

4.17.1 Medical Record Verification Queue

The interface shows an ordered list of students that have already passed through the academic procedures and are now ready for medical tests.

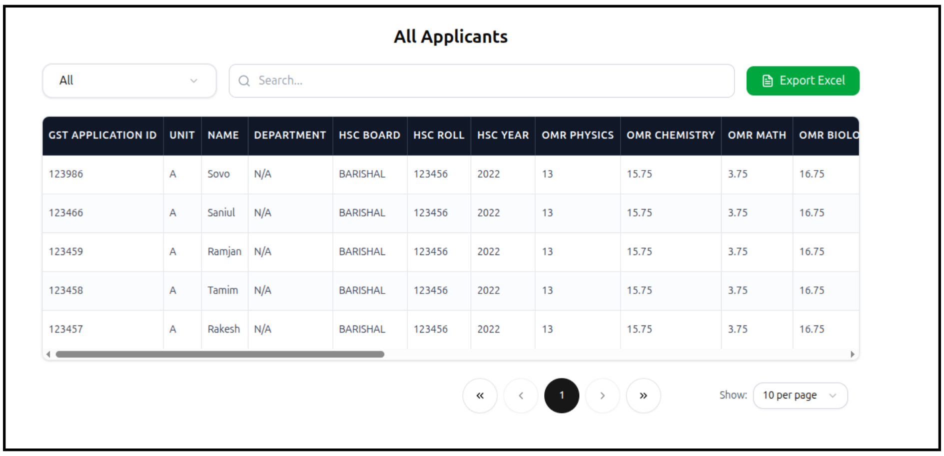


Figure 4.21: Medical Officer's Student Medical Record Verification Queue

4.17.2 Medical Record Detail Table Matrix

A specialized table structure is adopted to capture more information on the physical conditions of the student applicants.

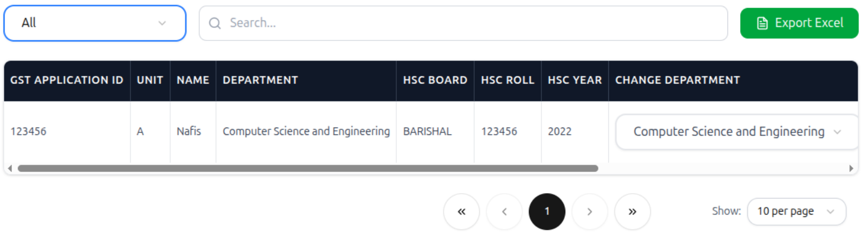


Figure 4.22: Medical Record Details Matrix

Chapter 5

Testing and Quality Assurance

5.1 Chapter Overview

This chapter will outline the detailed testing procedure carried out to guarantee the system's accuracy, security, and reliability. Initially, there was API testing where all the backend endpoints' accuracy of returning appropriate responses and handling the information were tested using Postman. Next, there was functional testing where all the functionalities of the system's interfaces were manually checked to ensure that all the form validations are working appropriately to provide users with a smooth experience when using the application. Security testing was one of the main tests carried out on the system where the role-based access control system was validated to ensure that unauthorized users cannot gain access to the application's administrative module.
To test the platform's reliability when subjected to a considerable number of requests, a load test was carried out where several requests were sent concurrently to test response times and stability. According to the results obtained from the various stages of testing, all the modules of the system worked perfectly without any challenges. Therefore, even after being loaded with many concurrent requests, the system showed reliability. Thus, the system's infrastructure for conducting an online admission process is fit for deployment.
The Digital Admission Procedure Management for Post-GST Candidates has been thoroughly tested to ensure that the system functions correctly and reliably. This stage is a crucial part of the development process before deploying the system for the 2026 admission session.

This chapter explains the different testing approaches used for the Undergraduate Final Admission System, including automated testing, backend API validation, and manual functional testing.

5.1.1 Utility Function Testing

We also tested important utility functions responsible for handling sensitive data such as dates and score calculations. All functions produced correct and expected outputs.

5.2 Backend API Testing with Postman

Postman was used to test the backend of the Undergraduate Final Admission System, which is developed using Node.js and Prisma. The backend successfully returned expected responses, confirming correct API functionality.

5.3 Manual Functional Testing

Manual testing was conducted to evaluate the system from a user perspective. The admission form was tested by submitting both valid and invalid inputs to verify system behavior.

5.3.1 Component Rendering Tests

Using Vitest and React Testing Library, we tested whether the main user interface components are rendering correctly. The results confirmed that the system UI elements, including titles and key action buttons, are correctly displayed and functional.

5.3.2 Form Validation Testing

All form fields were carefully tested to ensure proper validation. Invalid inputs were intentionally entered, and the system successfully detected and rejected incorrect data.

5.3.3 Role-Based Access Control (RBAC) Test

Attempts to access restricted administrative pages without proper authentication resulted in redirection to the login page, confirming secure access control implementation.

5.4 Load Testing and Performance Evaluation

The load testing of the PDF generation module was carried out to examine how the system operates while processing concurrent requests. All incoming requests were successfully processed during testing; no failed operations were recorded.

5.4.1 Outcomes

- 100% of the requests were successfully completed without any errors.
- Every API response was valid (HTTP 200).
- No system crashes, deadlocks, or disruptions were observed.
- The generated PDF outputs were consistent and accurate.

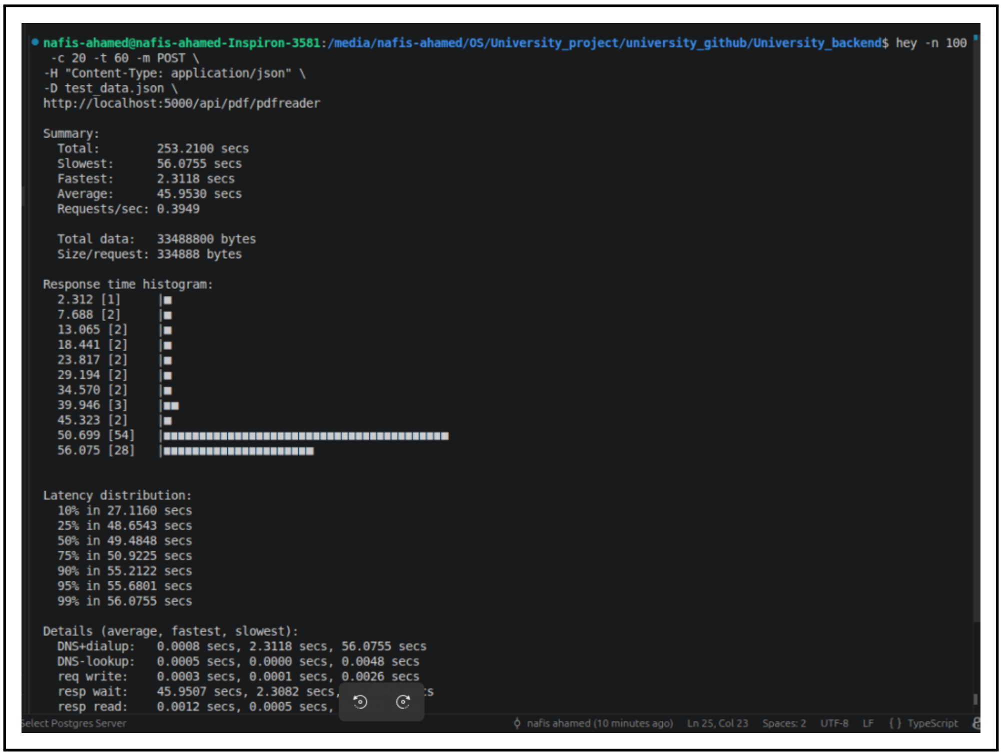


Figure 5.1: Load Testing Result Overview

5.5 Summary of Test Results

All testing phases were successful. The results of different testing categories are summarized in the table below.

Table 5.1: Summary of Testing Results for JUST-DAPM-PGC

Test Category	Testing Tool	Status
API Testing	Postman	Passed
Functional Testing	Manual	Passed
Validation	Manual	Passed
RBAC	Manual	Passed
Load Testing	Hey Tool	Passed

Chapter 6

Deployment

6.1 Chapter Overview

This chapter will be the culmination of the thesis with discussions on the implementation and final evaluation of the system. Implementation is the last stage in the software development life cycle and involves putting together the backend and frontend of the system on the university server so that the end-users can have the real and live experience using the Digital Admission Procedure Management system. By being able to deploy the Digital Admission Procedure Management system, we will have proven the viability of having an admission process made easier with automation.
The way forward from here is that the system will be successfully implemented. However, there will also be numerous areas in which the system can be improved in the future. There are still some improvements that the team would like to make to enhance functionality even more. These include the development of a mobile app and incorporation of artificial intelligence among others. Ultimately, the Digital Admission Procedure Management system will successfully digitize the admission process for Jashore University of Science and Technology.

6.2 Deployment Process

Deploying a web application built with Next.js and Node.js needs a structured workflow.

This ensures stability and performance.

The deployment process for DAPM-PGC followed these essential steps:

- Production Build: The application code was made optimized and compiled into a production build. This ensures fast loading times and efficient resource management.
- Environment Configuration: Essential environment variables like PostgreSQL database connection strings and Prisma client settings were configured on the production server.
- Server Setup: The application is hosted on the university's infrastructure. Due to existing resource allocations the system is currently deployed on a port.
- Monitoring: The system is actively maintained by the Technical Committee of Undergraduate Admission at Jashore University of Science and Technology.

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6.3 Deployment Status and URL

The Digital Admission Procedure Management for Post-GST Candidates is currently live.

It is accessible for the admission process.

- Hosting: The project is running on the university's existing domain using port 3000. This is a cost-effective hosting solution.

6.4 Future Work

While the current Digital Admission Procedure Management for Post-GST Candidates fulfills all requirements future updates could include:

- Automated Migration: Implementing an auto-migration system for students across different departments in the Digital Admission Procedure Management for Post-GST Candidates.
- Payment Gateway Integration: Integrating mobile banking like bKash and Nagad. This will enable admission fee collection for the Digital Admission Procedure Management for Post-GST Candidates.
- Mobile Application: Developing a mobile app. This will allow students to track their admission status in real time for the Digital Admission Procedure Management for Post-GST Candidates.

Through this project Jashore University of Science and Technology has taken a step forward.

Chapter 7

Conclusion

The deployment of the Digital Admission Procedure Management system marks the successful transition of the application from development to a real-world operational environment. Through proper configuration of the backend and frontend components, the system has been effectively hosted on the university infrastructure, ensuring accessibility for end-users.

This deployment demonstrates the practicality and reliability of the system in managing the admission process in a more efficient and automated manner. It reduces manual workload, improves data handling accuracy, and ensures smoother communication between applicants and the administrative office.

Although the system is currently fully functional, there is still room for future enhancements such as mobile application support, improved scalability, and integration of advanced technologies like artificial intelligence. These improvements can further increase usability and performance.

Overall, the deployment phase confirms that the Digital Admission Procedure Management system is a successful step toward digitalizing the admission process at Jashore University of Science and Technology.

Another key aspect brought out by the implementation process is the role of cooperation between the development team and the administrative body of the institution. It ensures that the software application fulfills all institutional needs and reflects the actual admission process.On another note, the software application has been created to be scalable and thus capable of being expanded without any need to redesign.

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